Version: 30092023

Big Data Fundamentals

Overview

Data Science is one of the most in-demand skillsets that companies are hiring for. No matter you are looking for becoming a data analyst, data engineer or data scientist, they all share the same core skillset - working with data. Establishing how to think about data will set you up for success.

In this course, you will build the conceptual foundation you need to work with data and present your insights. At the end of the course, you will be capable of articulating what can go right and what can go wrong when working with data.

Unit 1
Data Literacy

What will you learn

- Basic assumptions of working with different types of data
- How data types affect the analysis
- Foundational statistical concepts
- What is good data visualizations

Data Gaps

- Our data-driven conclusions are only as strong, robust, and well-supported as the data behind them

- Imagine you have a model that can accurately predict the weather in Hong Kong 90% of the time, but then using data from only Tokyo weather stations as the input values. Or using temperature data that was collected only at noon each day.

- No matter how good the model is, it will need accurate data to give a helpful output

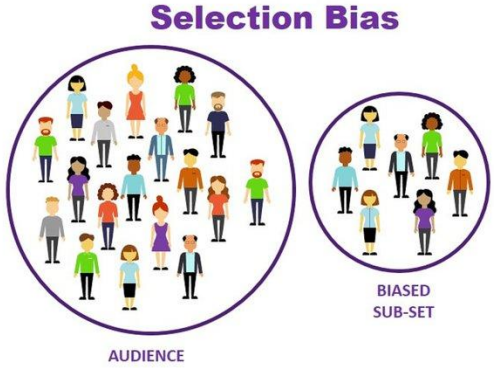
- Practicing good data literacy means asking:
  - Do we have sufficient data to answer the question at hand?
  - Can my data answer my exact question?

Addressing Bias

- If your data contains bias, the model will learn them and reflect those biases in the result

- For example, a study by US National Institute of Standards and Technology in 2018 found that some widely used facial recognition algorithms had significantly higher error rates when identifying people with darker skin tones and women. Can you imagine why?

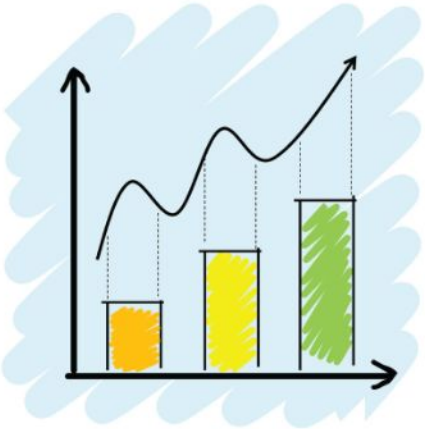
- Practicing good data literacy means asking:
  - Who participated in the data?
  - Who is left out?
  - Who made the data?



Statistics At Work

- Statistics can reveal systemic patterns in a data set rather than relying on individual experiences

- We can analyze data, make inferences, and draw conclusions with statistical technique. Through statistical methods such as hypothesis testing, confidence intervals, and regression analysis, we can assess the reliability of our findings and make data-informed decisions

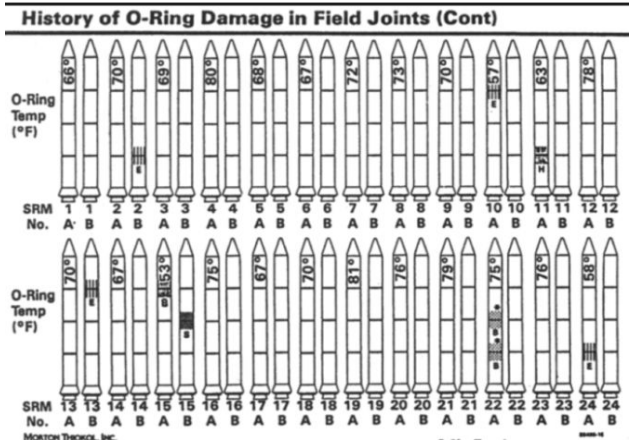


Visualization is important

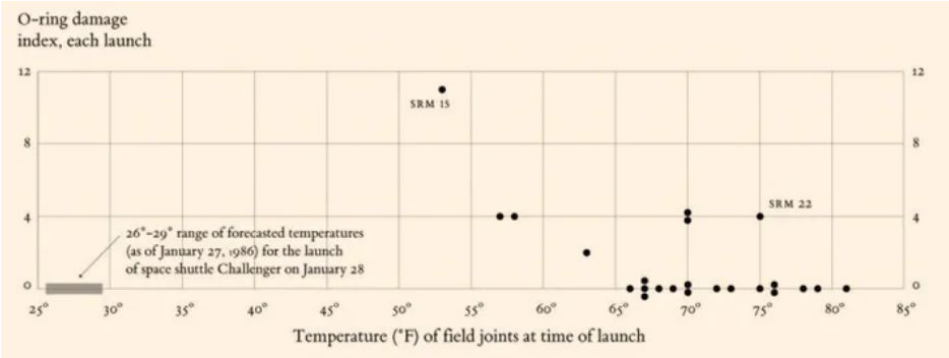
- Data visualization helps us to explore and understand data-driven arguments and is a powerful tool for communication

- In 1986, the Space Shuttle Challenger broke apart 73 seconds into its flight, killing all seven crew members aboard. The explosion was caused by a failure of two O-rings. Before launched, the engineers proposed to postpone the launch using a chart that showed launch success rates at various temperatures, but their arguments did not prevent the launch

Original chart: Can you see the relationship?



Revised chart later by statistician

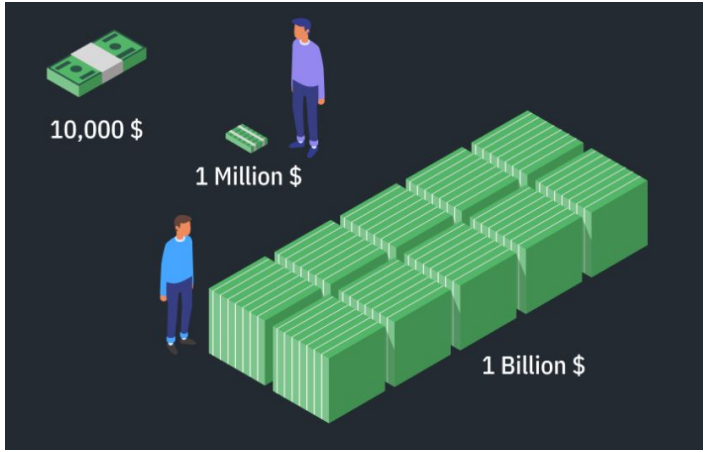


Numeracy

- We're generally good at numbers we see everyday, for example, the amount to purchase a cup of coffee or pay a utility bill. We can visualize what those numbers mean and understand the effect of them increasing or decreasing by certain percentage, like 10%

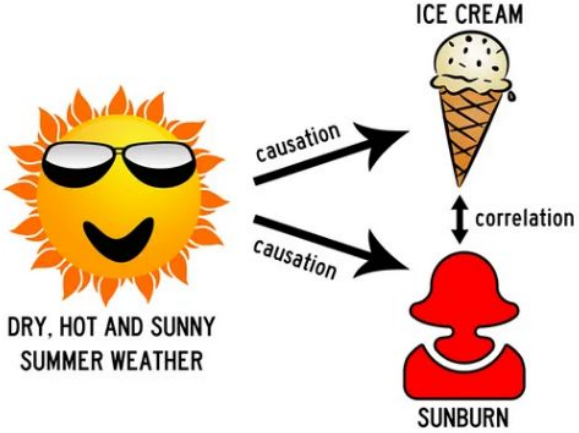
- But numbers without common usage are another story. Think about the GDP of a country, if it changes by 10%, what does the change actually mean in real life terms?

- Part of the data science job is to provide context and clarifications to make sure that audiences are not only reading the numbers, but understanding what's the takeaway



Correlation does not equal Causation

- Two events might be connected or related, that doesn't mean they're in a cause-and-effect relationship
- Correlation means there is a relationship or pattern between the values of two variables
- Causation means that one event causes another event to occur. Causation can only be determined from an appropriately designed experiment.



Data Ethics and Privacy

- Most people are aware that data visualizations, machine learning algorithms, and analyses require data. But where does that data come from? Does everyone have access to data for analysis?

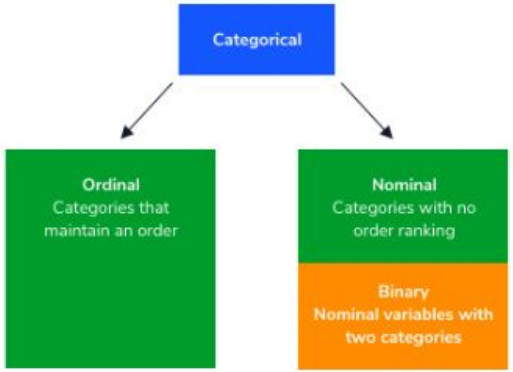
- Much of the data available to us comes from individuals and would be considered personally identifiable, meaning we could use it to identify someone. Ethical issues regarding data collection:
  - **Consent**: Individuals must be informed and give their consent for information to be collected
  - **Ownership**: Anyone collecting data must be aware that individuals have ownership over their information
  - **Intention**: Individuals must be informed about what information will be taken, how it will be stored, and how it will be used
  - **Privacy**: Information about individuals must be kept secure

Variable Types

- Variables that are measured are **Numerical variables.** Variables that are categorized are **Categorical variables**

- There are two ways to get a **numerical number**: by counting and measuring
  - Counting gives us whole numbers and **discrete** variables
  - Measuring gives us potentially partial values and **continuous** variables

- **Categorical variables** describe characteristics with words or relative values
  - A **nominal variable** means a named value (e.g. teacher name)
  - A **binary variable** has only two options: Yes and No
  - Subjectively say on a scale of 1 to 5 is called an **ordinal variable**



Data Shapes

- In a well-organized dataset, the variables (e.g. teacher_name) describe a characteristic of the entities (e.g. teacher). Good variables measure only one characteristic and should not be a characteristic themselves

Bad Data Shapes:

teacher_id	teacher_name	bachelor_degree	master_degree	doctor_degree
1	Helen	CUHK	nan	nan
2	Brenda	nan	HKU	nan
3	Jason	nan	nan	UCLA

Good Data Shapes:

teacher_id	teacher_name	Education Level	University
1	Helen	bachelor	CUHK
2	Brenda	master	HKU
3	Jason	doctor	UCLA

Missing data

- If the data just wasn't entered properly. This kind of missing is **Missing Completely at Random** (e.g. You removed 10% of the respondents data randomly)

- If one value is missing because of some random characteristic about the person or thing being studied. This kind of missing is called **Missing at Random**
(e.g. People who come from poorer families might be less inclined to answer questions about drug use, and so the level of drug use is related to family income)

- If the data can be structurally missing, meaning that we wouldn't expect a value there to begin with. This kind of missing is **Structurally Missing**
(e.g. Students skipped the question on drug use because they feared that they would be expelled from school)

- For structurally missing data, we can just ignore it. For Missing at Random and Missing Completely at Random, there is an entire science behind what to do with these values

Correct data

- Can you spot the error and correct it?

teacher_id	teacher_name	Industry Exam Score	Education Level	University
1	Helen	79.3	bachelordegree	CUHK
2	Brenda	90.5	masterdegree	HKU
3	Jason	82.7	doctor degree	UCLA
4	Nathan	8.53	bachelordegree	cuhk
5	Jenny	93.2	masterdegree	HKU

Data Accuracy

To evaluate the accuracy of dataset:

1. Thinking about the data against expectations and common sense is crucial for spotting issues with accuracy. You can do this by **inspecting the distribution and outliers** to get clues about what the data looks like

2. Considering **how error could have crept in during the data collection process** will help you group and evaluate the data to uncover systematic inconsistencies

3. Identifying ways that **duplicate values** could have been created

Data Validity

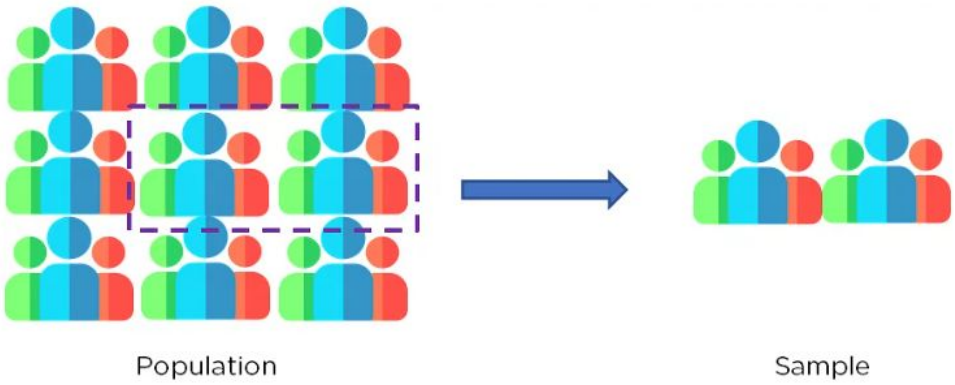
- We also have to make sure that our data actually measures what we think it is measuring. This is the **validity** of our dataset

- For example, if we decide that industry exam score and education level are related, and we use education level as a proxy for teacher quality. With that decision, we just compromised the validity of our dataset

- The best way to spot issues with the validity of a dataset is to ask: **Does this variable measure what I think it does**?

Representative Samples

- The sample should look like the population in as many characteristics as possible. There are a lot of techniques for creating representative samples, but they all have the same goal: **to find a mix of observations that contains all of the features in the larger population**



Describing Categorical Variables

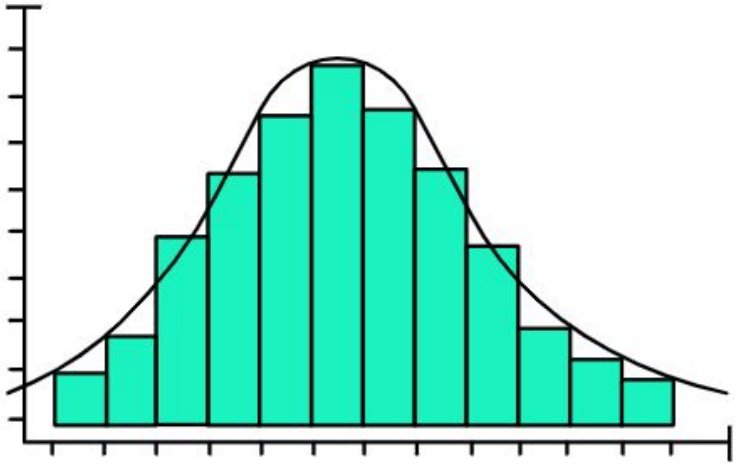
- **Frequency**: the count of teachers for each job title
- **Proportion**: the frequency divided by the total number of teachers
- **Percentage**: the proportion converted from a decimal to a percentage

Title	Frequency	Proportion	Percentage
Junior teacher	350	0.44	44%
Associate teacher	113	0.14	14%
Subject teacher	87	0.11	11%
Senior teacher	239	0.30	30%
Total	789	1	100%

Describing Numeric Variables

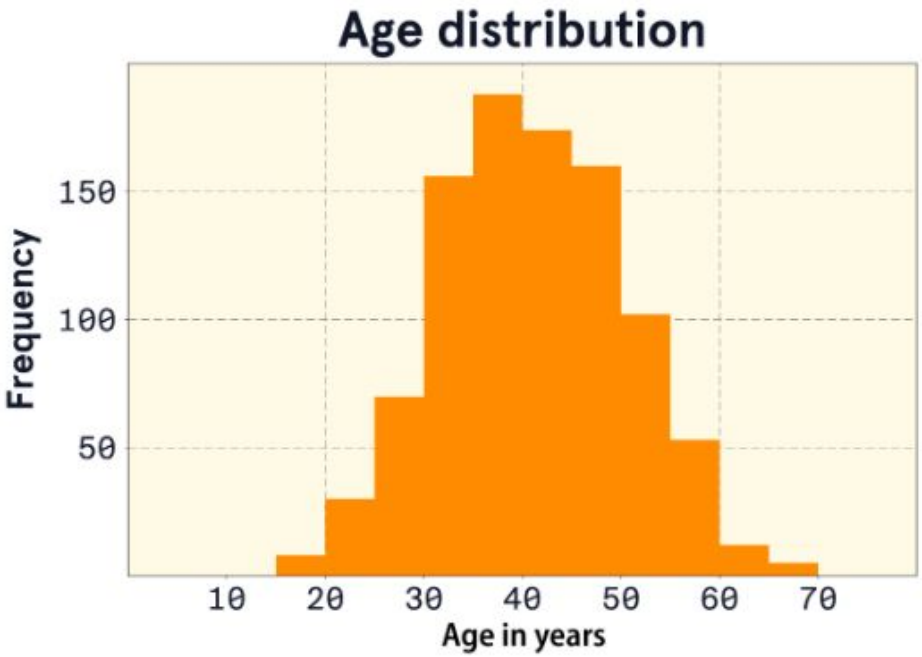
- There are a lot of ways we can describe the **distribution** of a numeric variable. A distribution is a function that shows all possible values of a variable and how frequently each value occurs

- This distribution might be considered bell-shaped. This is actually a very common pattern and is called a **normal distribution**



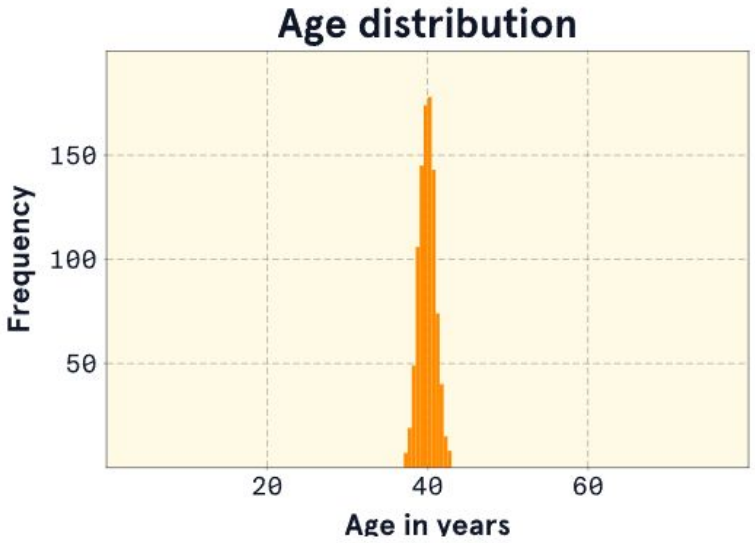
Describing Numeric Variables

- What would you say is the typical age of a teacher in Hong Kong?
- Are most teachers about this age, or are there lots of teachers of many different ages?



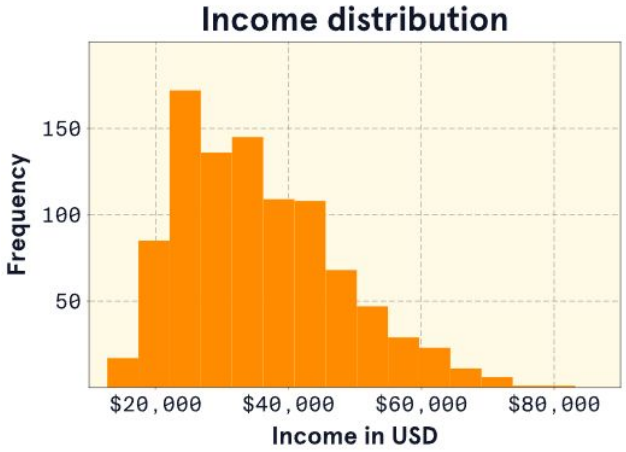
Mean and Standard Deviation

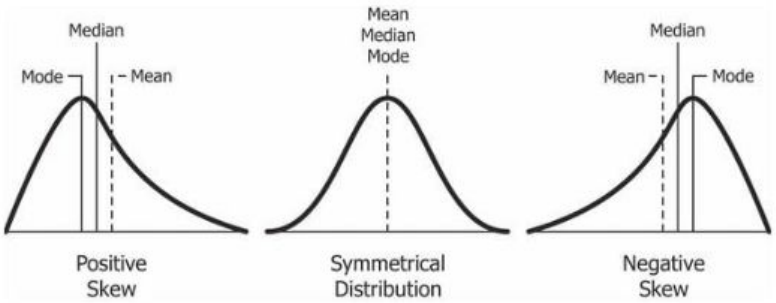
- The **mean**, also called the average, describes the **center** of a **numeric distribution** by adding all values and dividing by the count.
- The **standard deviation** describes **the spread of values in a numeric distribution relative to the mean**. It is calculated by finding the average squared distance from each data point to the mean and square-rooting the result.



Skewed Distributions

- We notice that the shape of the distribution is different than the shape of the age distribution. There are quite a few teachers with higher incomes that are creating a longer tail on the right side
- What we have learned is that the income distribution is **skewed**. A skewed distribution is asymmetrical. Specifically, the income distribution is **right-skewed** (also called positively-skewed) because the tail is on the right side





Median

- We want to find a value that represents the typical income, but we don't want to use the mean because the data are skewed
- One method would be to find the middle value when all values are arranged from smallest to largest. This value is called the **median**, it's also referred to as the **50th percentile**
- The median is the best measure of central tendency because it's the value exactly in the middle

1, 3, 3, **6**, 7, 8, 9

Median = **6**

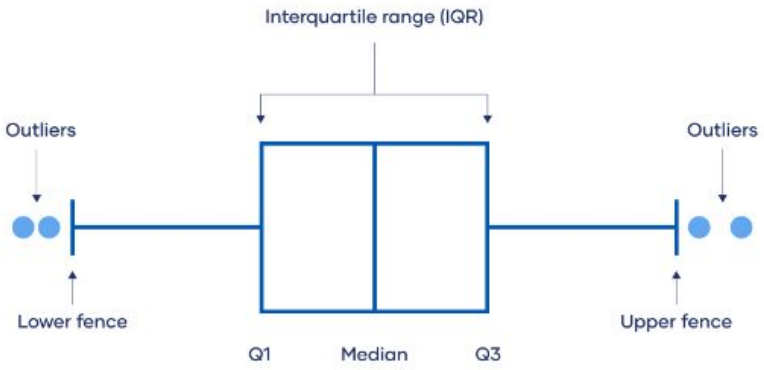
1, 2, 3, **4**, **5**, 6, 8, 9

Median = $(4 + 5) \div 2$

= **4.5**

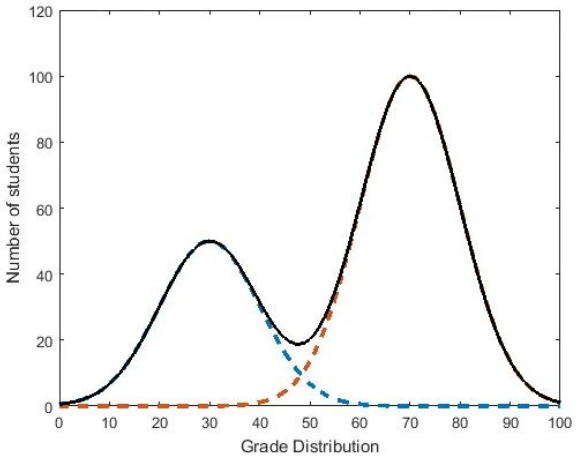
The interquartile range (IQR)

- The **interquartile range** is an especially useful measure of variability for skewed distributions
- The interquartile range is found by subtracting the Q1 value (25th percentile) from the Q3 value (75th percentile), marking the range for just the middle 50% of the data
- Just as with skewness, **outliers** tend to more heavily influence the mean than the median. This same pattern occurs with measures of spread: the standard deviation is more influenced by outliers and skewness than the interquartile range



Mode

- The mode is defined as the value with the highest frequency, but we can also think of the mode as the value where the peak of the distribution occurs. While not great for computations, the mode can help us identify interesting features in a variable.



Variable Relationships

- We specifically looked at relationships between two numeric variables with **scatter plot**
- We can describe the relationship more precisely by measuring the **correlation coefficient**. This number ranges from -1 to +1 and tells us two things about a linear relationship:
- **Direction**: A positive coefficient means that higher values in one variable are associated with higher values in the other
- **Strength:** The farther the coefficient is from 0, the stronger the relationship and the more the points in a scatter plot look like a line

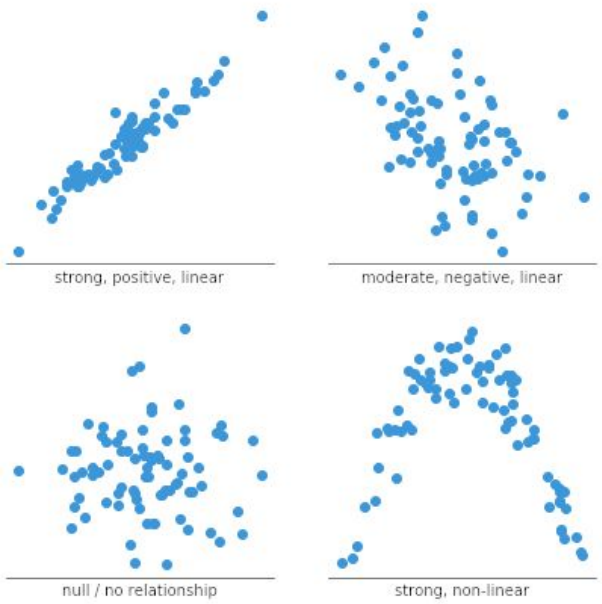
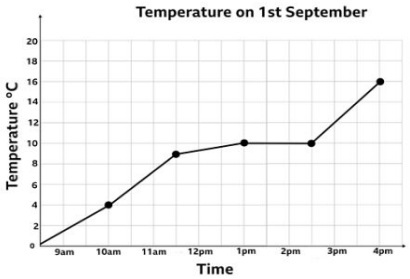
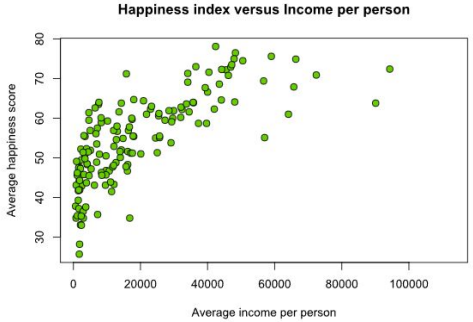
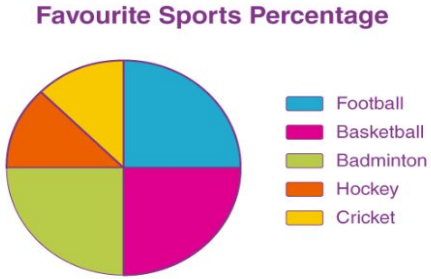


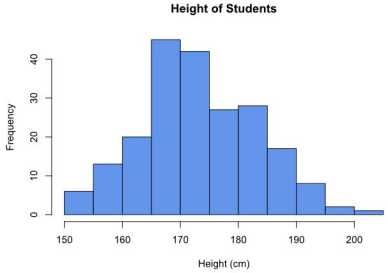
Chart Type

- The first step of making a data visualization is choosing a chart type, which is important for communicating about the relationship we want to show



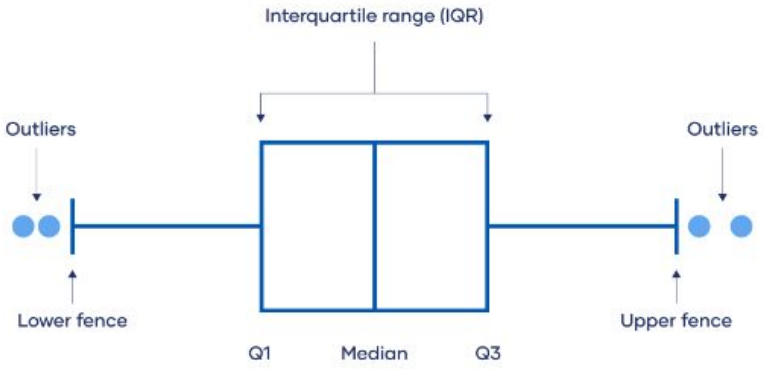






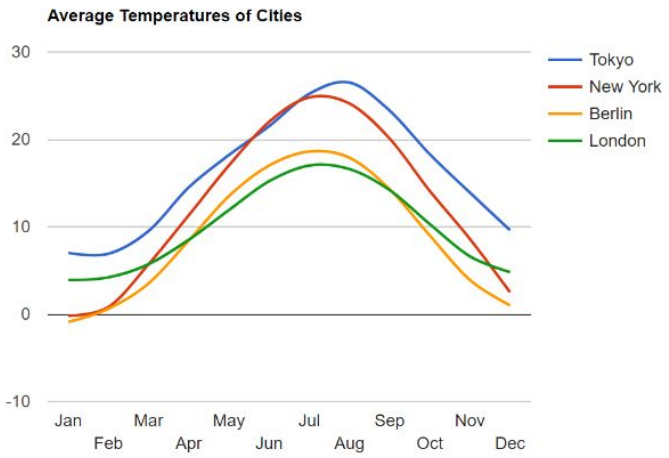
Univariate Charts

- One big consideration when choosing a chart type is **how many variables we're comparing**. **Univariate charts** help us visualize a change in one variable
- If we want to compare an **amount between different categories** - **bar chart**
- **Histograms** measure the **distribution**, or spread, of a variable
- A more "math-forward" way to visualize distributions is a **box plot** with percentile



Bi- and Multivariate Charts

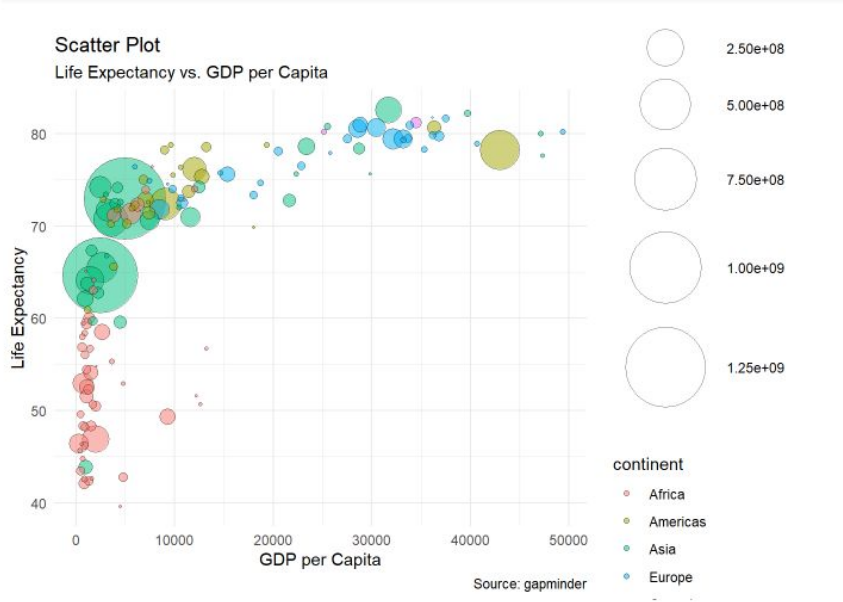
- **Multivariate Charts** show the relationships between **two or more variables**
- A **scatter plot** - one variable on the x-axis, another on the y-axis, and each point helps us compare the two variables by its position on the graph. It only makes sense for numeric variables, not categorical
- A **line chart** with multiple lines for different variables is a multivariate chart



Aesthetic Properties

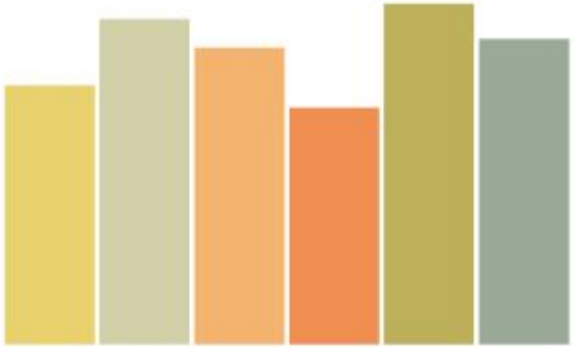
Aesthetic properties are the attributes help us to communicate data, including:

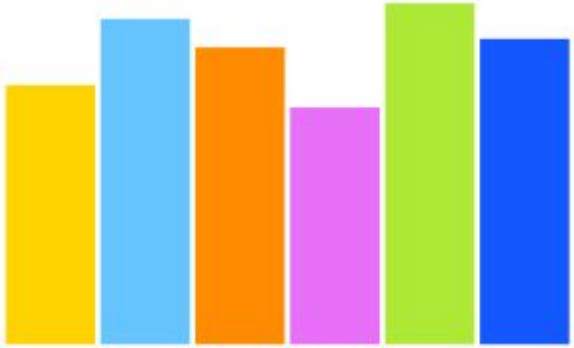
- Position
- Size
- Shape
- Color / pattern



Color Accessibility

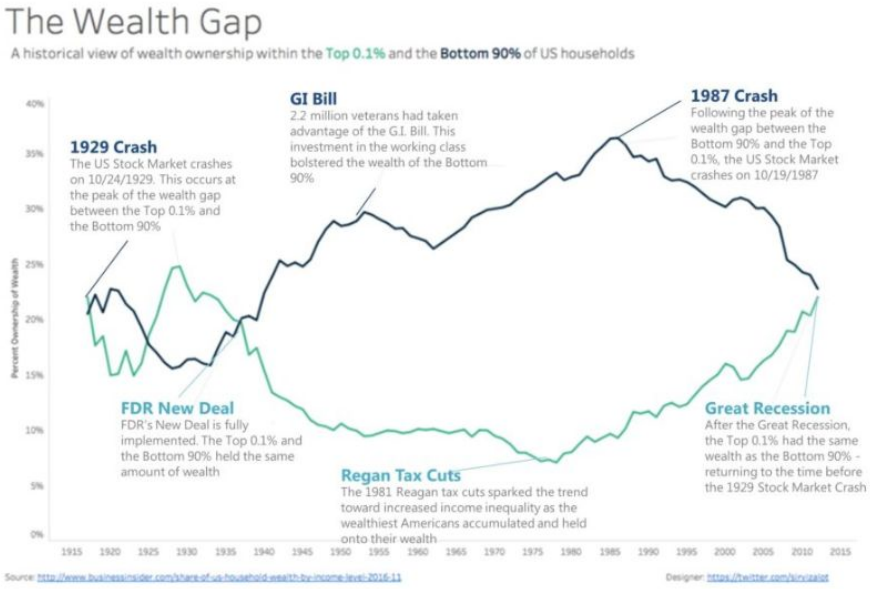
- Think not only about the hue of a color (e.g. red, green, or purple), but the value as well (e.g. bright red, light green, dark purple). Good color comparisons use high contrast values, not just different hues
- Large enough font size!





Consider the Audience

- Different context is appropriate for different audiences
- Personalize the chart with a title and annotations that work best for the intended audience
- Considering specific audiences when making decisions about what context to include or not

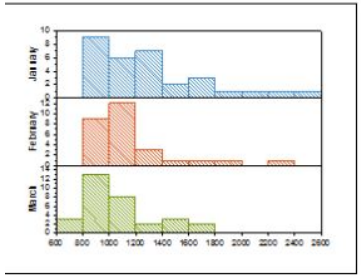


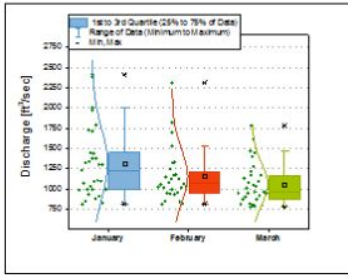
Descriptive Analysis

- Descriptive analysis lets us describe, summarize, and visualize data so that patterns can emerge. Most of the time a descriptive analysis is the first step in our analysis process because it can draw basic conclusions about the data at hand
- Descriptive analyses include measures of central tendency (e.g., mean, median, mode) and spread (e.g., range, quartiles, variance, standard deviation, distribution), which are referred to as descriptives or summary statistics

Descriptive Statistics

	N total	N missing	Mean	Standard Deviation	Sum	Minimum	Median	Maximum
January	31	1	1315.35	437.33	40776	814	1230	2410
February	28	0	1158.21	332.80	32430	816	1065	2310
March	31	2	1061.52	258.58	32907	779	970	1780

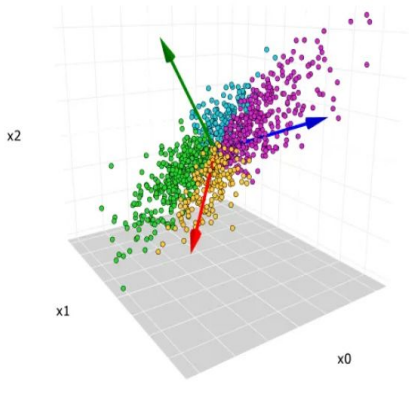




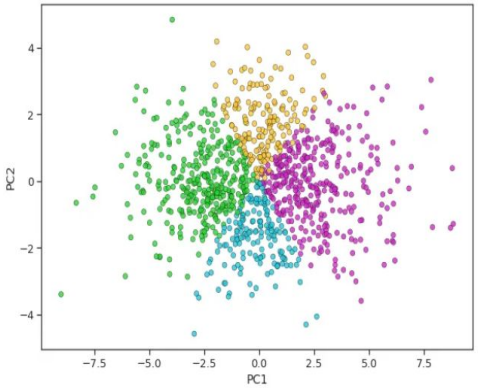
Exploratory Analysis

- Exploratory analysis is the next step after descriptive analysis. With exploratory analysis, we **look for relationships between variables** in our dataset.
- Unsupervised machine learning techniques, such as **Clustering** algorithms, are useful tools for exploratory analysis. These techniques “learn” patterns from untagged data and they help us see relationships between many variables at once
- We can perform a **Principal Component Analysis** or PCA, which compresses the variables into principal components that can be plotted against each other. After PCA, we can use k-means clustering to look for trends in the data.

Before PCA

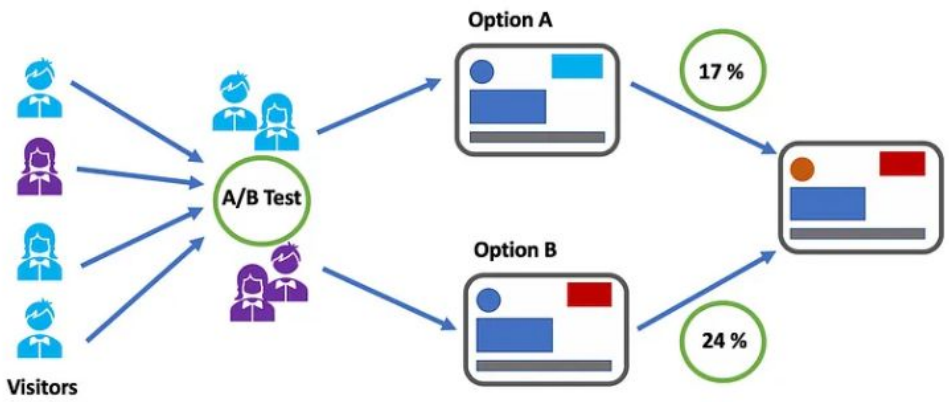


After PCA



Inferential Analysis

- Inferential analysis lets us test **a hypothesis** on a sample of a population and then extend our conclusions to the whole population. A/B tests are a type of **inferential analysis**.
- Sample size must be big enough (10% is a good rule-of-thumb)
- Sample must be randomly selected and representative of the total population
- We can only test one hypothesis at a time



Causal Analysis

- Causal analysis generally relies on carefully designed experiments, but we can sometimes also do causal analysis with observational data.

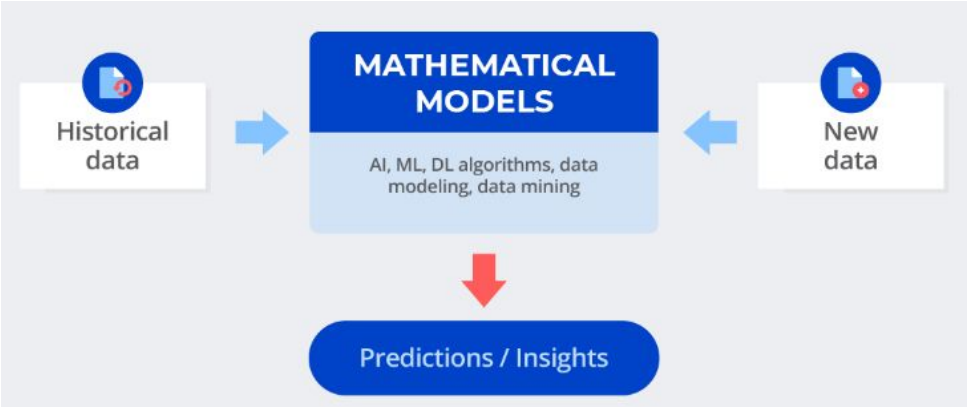
- Only change one variable at a time and carefully control all other variables

- For example, could you determine if drinking more water caused better sleep by monitoring your own sleep for a month and comparing results from days that you drank 8 glasses of water and days that you did not?

- Could you determine if drinking more water caused better sleep if you expanded your study to include 1,000 subjects and asked all men to record their sleep on days they drank 8 glasses of water and all women to record their sleep on days that they did not drink 8 glasses of water?

Predictive Analysis

- **Predictive analysis** uses data and supervised machine learning techniques to identify the likelihood of future outcomes
- Some popular supervised machine learning techniques include regression model and deep learning neural networks. The actual algorithm used with each of these techniques is different, but each requires **training data**



Recognize bias

- **Selection bias**
  Study subjects (i.e., the sample) are not representative of the population

- **Algorithmic bias**
  Algorithm produces systematic and repeatable errors that lead to unfair outcomes

- **Confirmation bias**
  Tendency to seek out information that supports our views

- **Overgeneralization bias**
  Inappropriately extending observations made with one dataset to other datasets

- **Reporting bias**
  Only report or share results that affirm our beliefs or hypotheses

Unit 2
Statistics Fundamentals

What will you learn

- Basic probability distribution
- Sampling distribution and simulating binomial test
- Hypothesis testing and significance thresholds
- Introduction to linear regression

Introduction to Probability

- Probability allows us to measure uncertainty
- For example, when you check the weather on your phone, you are using probability and intuition to gauge what you will wear that day

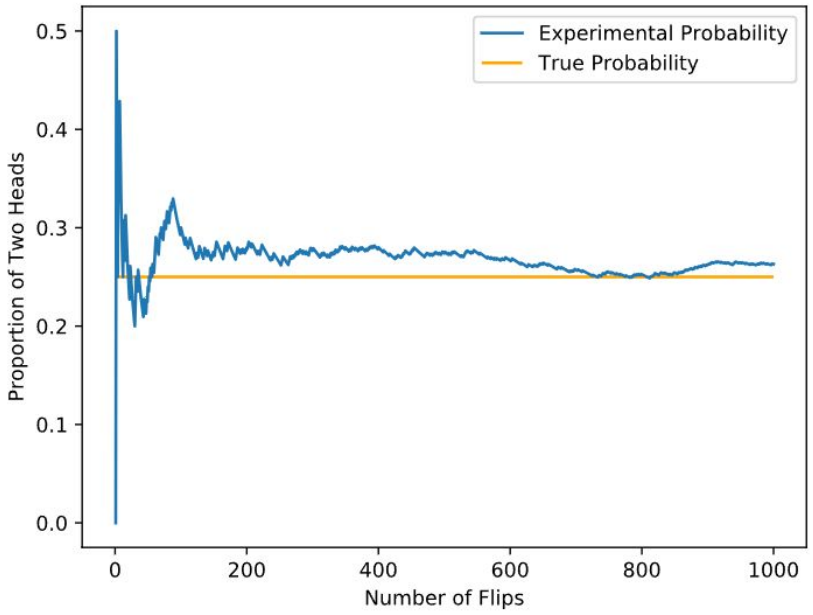
Experiments and Sample Spaces

- An experiment is something that produces observation(s) with some level of uncertainty
- A sample point is a single possible outcome of an experiment
- A sample space is the set of all possible sample points for an experiment
- Flip two coins once, the sample space is:
  - S = {HH, TT, HT, TH}
- Probability of event = P(event) = # of times event occurred / total number of trials
- For example, if you flipped two coins 1000 times and get 300 times HH:
  - P(HH) = 300 / 1000 = 0.3
- However, if you do this over and over again, you may get slightly different results

Law of Large Numbers

- As we flip two coins more and more, the observed proportion of times each event occurs will converge to its true probability

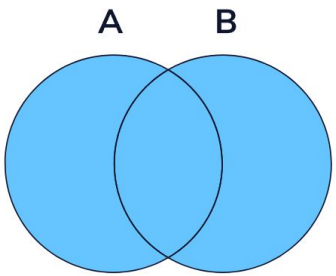
```python
import matplotlib.pyplot as plt
import numpy as np

def coin_flip_experiment():
  # defining our two coins as lists
  coin1 = ['Heads', 'Tails']
  coin2 = ['Heads', 'Tails']

  # "flipping" both coins randomly
  coin1_result = np.random.choice(coin1)
  coin2_result = np.random.choice(coin2)

  # checking if both flips are heads
  if coin1_result == 'Heads' and coin2_result == 'Heads':
    return 1
  else:
    return 0

# how many times we run the experiment
num_trials = 1000
prop = []
flips = []
# keep track of the number of times heads pops up twice
two_heads_counter = 0

# perform the experiment five times
for flip in range(num_trials):
  # if both coins are heads add 1 to the counter
  two_heads_counter += coin_flip_experiment()
  # keep track of the proportion of two heads at each flip
  prop.append(two_heads_counter/(flip+1))
  # keep a list for number of flips
  flips.append(flip+1)

# plot all flips and proportion of two heads
plt.plot(flips, prop, label='Experimental Probability')
plt.xlabel('Number of Flips')
plt.ylabel('Proportion of Two Heads')

plt.hlines(0.25, 0, num_trials, colors='orange', label='True Probability')
plt.legend()
```



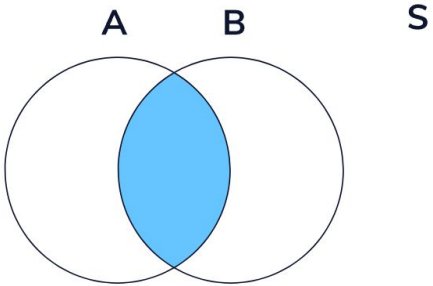
Rules of probability - Union, Intersection & Complement

- The union of two sets encompasses any element that exists in either one or both of them
- The intersection of two sets encompasses any element that exists in both of the sets.
- The complement of a set consists of all possible outcomes outside of the set

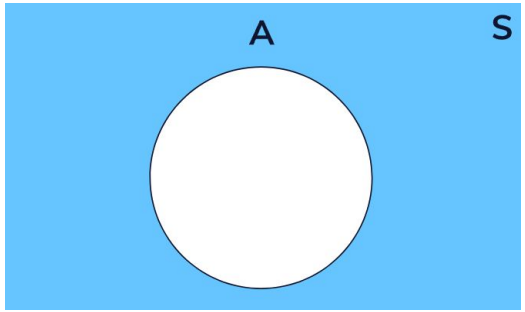
Union



Intersection



Complement

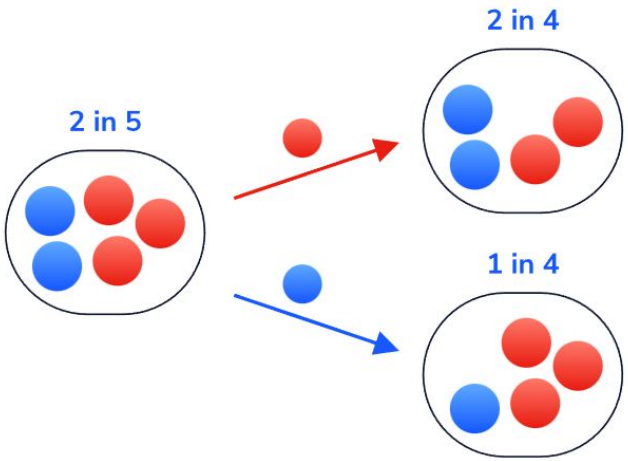


Independence and Dependence

- Imagine that we flip a fair coin 5 times and get 5 heads in a row. Does this affect the probability of getting heads on the next flip?

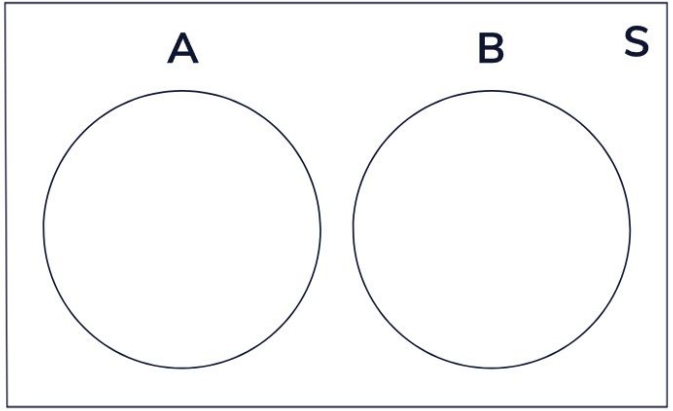
- Suppose we have a bag of five marbles: two marbles are blue and three marbles are red. If we take one marble out of the bag, what is the probability that the second marble take out is blue?

- What if we had put back the first marble? Is the probability that we pick a blue marble second independent or dependent on what we pick out first?



Mutually Exclusive Events

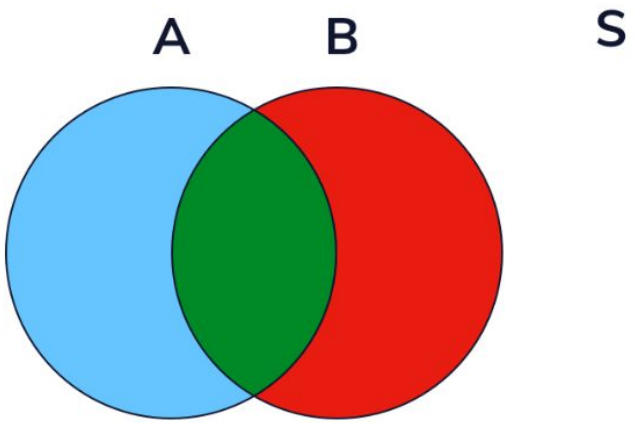
- Two events are considered mutually exclusive if they cannot occur at the same time



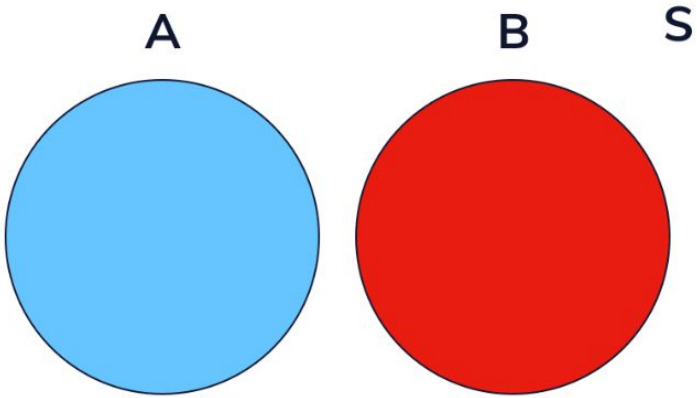
Addition Rule

- Event A is rolling an odd number on a six-sided die and Event B is rolling a number greater than two. What if we want to find the probability of one or both events occurring?
  - P (A or B) = P(A) + P(B) - P (A and B) if the events are not mutually exclusive
  - P (A or B) = P(A) + P(B) if the events are mutually exclusive

<u>Not mutually exclusive</u>

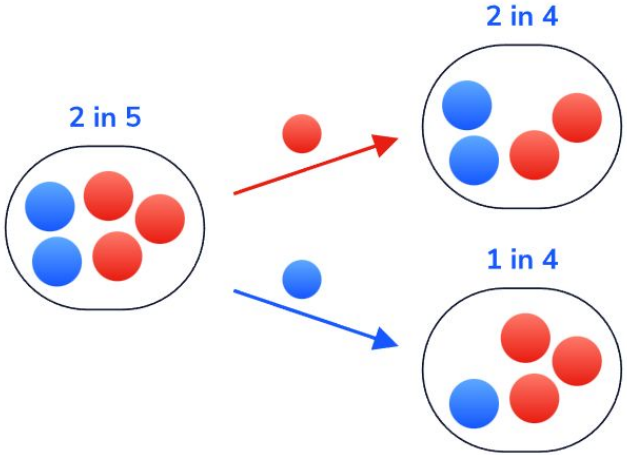


<u>Mutually exclusive</u>



Conditional Probability

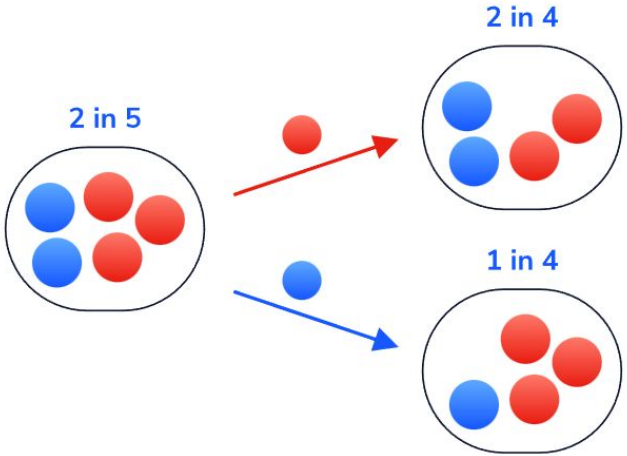
- Conditional probability measures the probability of one event occurring, given that another one has already occurred
- If we pick two marbles from a bag of five marbles without replacement, the probability that the second marble is red depends on the color of the first marble
- P (Red second | Blue First) = ¾
- For independent events: P(A|B) = P(A) and P(B|A) = P(B)



Multiplication Rule

- We have five marbles: two are blue, and three are red. We pick two marbles without replacement. What if we want to know the probability of choosing a blue marble first AND a blue marble second?
  - P(A and B) = P(A) * P(B|A) if two events are dependent
  - P(A and B) = P(A) * P(B) if two events are independent

- P(blue 1st and blue 2nd) = ⅖*¼ = 1/10



Exercise - Conditional Probability

- Suppose that the following is true :
  - 80 percent of the population has short-sighted: P(SS) = 0.8
  - 20 percent of the population does not have short-sighted: P(Not SS) = 0.2
  - If a person has short-sighted, there is an 85% chance their test will be positive and a 15% chance it will be negative: P(+|SS) = 0.85 and P(-|SS) = 0.15
  - If a person does not have short-signed, there is a 98% chance their test will be negative and a 2% chance it will be positive: P(+|Not SS) = 0.02 and P(-|Not SS) = 0.98

- What is the probability that a person has short-sighted and test positive? P(SS and +)
  - P(SS and +) = P(SS)*P(+|SS) = 0.8*0.85 = 0.68

- What is the probability that a person has short-sighted given a positive test result? P(SS | +)

Bayes' Theorem

- $P(B|A) = P(A|B) * P(B) / P(A)$

- In our example:
  - $P(SS|+) = P(+|SS)*P(SS)/P(+) = 0.85*0.8/0.684 = 0.9941$
  - $P(+) = P(+ \text{ and } SS) + P(+ \text{ and Not } SS) = 0.68 + 0.004 = 0.684$
  - $P(\text{Not } SS \text{ and } +) = P(\text{Not } SS)*P(+|\text{Not } SS) = 0.2*0.02 = 0.004$

Random Variables

- In probability, we often use random variables to represent random events
- For example, we could use a random variable to represent the outcome of a die roll: any number between one and six
- Random variables with a countable number of possible values are called **discrete random variables**. For example, rolling a regular 6-sided die would be considered a discrete random variable
- When the possible values of a random variable are uncountable, it is called a **continuous random variable**. These are generally measurement variables and are uncountable because measurements can always be more precise, for example, the temperature in Hong Kong on a randomly chosen day is a continuous random variable

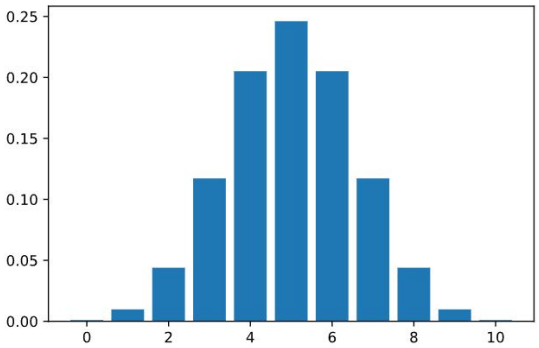
Probability Mass Functions

- A probability mass function (PMF) is a type of probability distribution that defines the probability of observing a particular value of a discrete random variable. For example, a PMF can be used to calculate the probability of rolling a three on a fair six-sided die

- The probabilities for all possible values in a given probability distribution add up to 1

- The probability mass function that describes the likelihood of each possible outcome (eg., 0 heads, 1 head, 2 heads, etc.) is called the **binomial distribution**

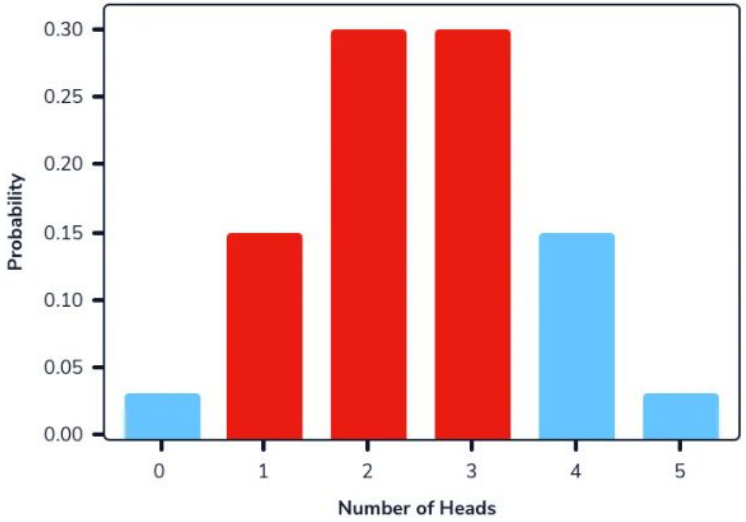
- For example, suppose that we flip a fair coin some number of times and count the number of heads. The probability mass function that describes the likelihood of each possible outcome (eg., 0 heads, 1 head, 2 heads, etc.) is called the binomial distribution



Using the Probability Mass Function Over a Range

- What if we want to find the probability of observing a range of values for a discrete random variable?
- We could do this is by adding up the probability of each value
- For example:
  P(1 to 3 heads) =
  P(1<=X<=3) =
  P(X=1)+P(X=2)+P(X=3)

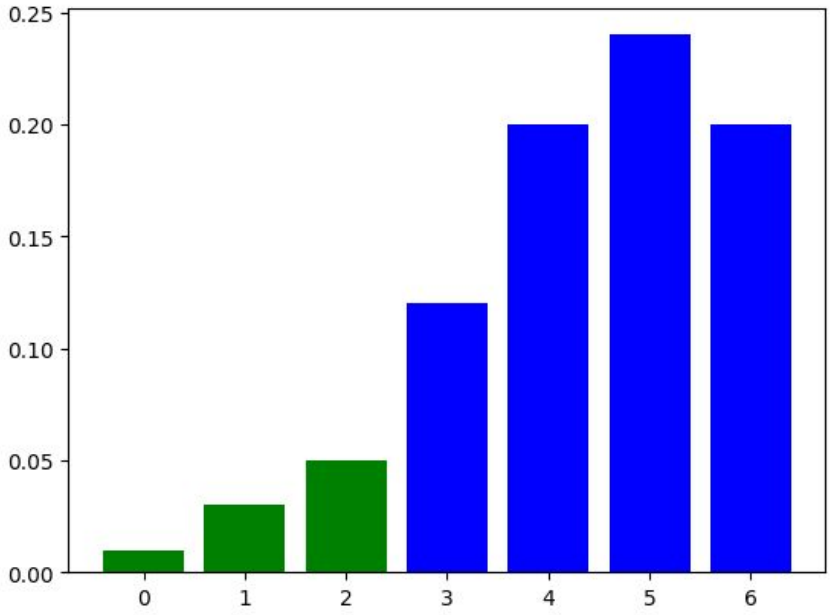
  P(2 to 5 heads) =
  P(2<=X<=5) =
  1 - P(0 head) - P(1 head) - P(2 head)



P(1, 2, or 3 heads) = P(X=1) + P(X=2) + P(X=3)

Cumulative Distribution Function

- To find the probability of observing between 3 and 6 heads, we can take the probability of observing 6 or fewer heads and subtracting the probability of observing 2 or fewer heads. This leaves a remnant of between 3 and 6 heads:
  - P (3<=X<=6) = P(X<=6) - P(X<=2)



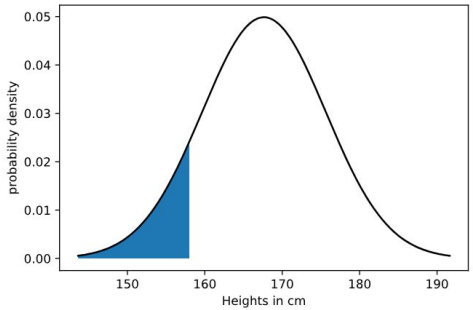
Probability Density Functions

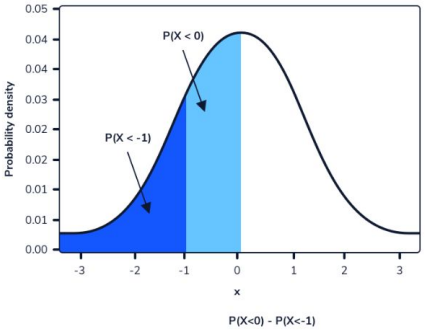
- Probability density functions define the probability distributions of continuous random variables and span across all possible values that the given random variable can take on

- We cannot calculate the probability at a single point. This is because the area of the curve underneath a single point is always zero

- We can calculate the area under the curve using the cumulative distribution function for the given probability distribution

- We can calculate the probability of observing these values or less. The difference between these two probabilities will be the probability of randomly observing a value in this given range.



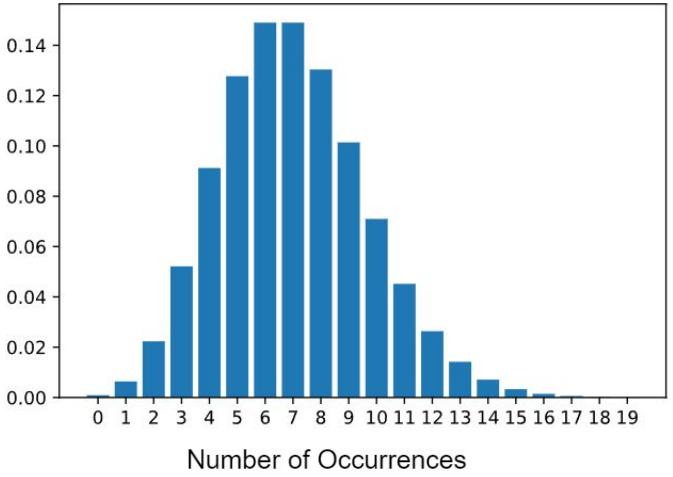


Poisson Distribution

- The Poisson distribution is another common distribution, and it is used to describe the number of times a certain event occurs within a fixed time or space interval

- For example, the Poisson distribution can be used to describe the number of cars that pass through a specific intersection between 4pm and 5pm on a given day

- The Poisson distribution is defined by the rate parameter, lambda: λ



Calculating Probabilities of a Range

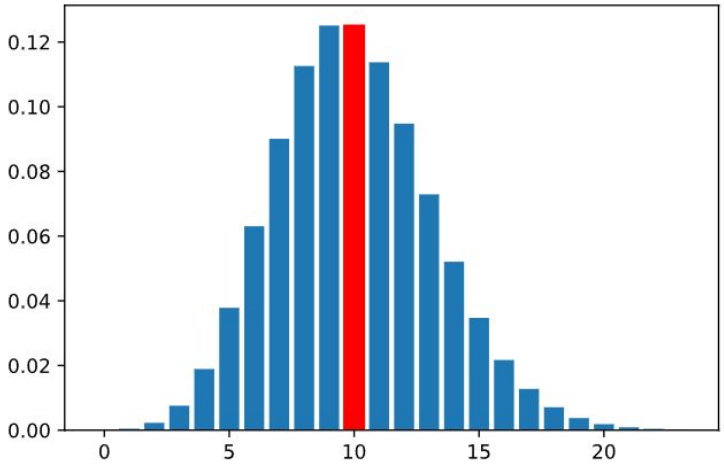
- If we wanted to calculate the probability of observing 12 or more rain events in the next 30 days when we expected 10, we could do the following:
  - CDF Poisson(12 or more,10) = 1 - CDF Poisson(11,10)

- Note that we used 11 above even though 12 was the value given in the prompt. We wanted to calculate the probability of observing 12 or more rains, which includes 12, so 1 - CDF Poisson(11, 10) would equal the probability of observing 12 or more rains

Expectation of the Poisson Distribution

- Let's say we are salespeople, and after many weeks of work, we calculate our average to be 10 sales per week. If we take this value to be our expected value of a Poisson Distribution, the probability mass function will look as follows

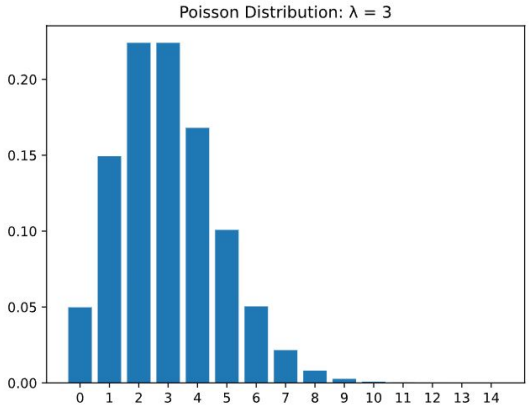
- The tallest bar represents the value with the highest probability of occurring

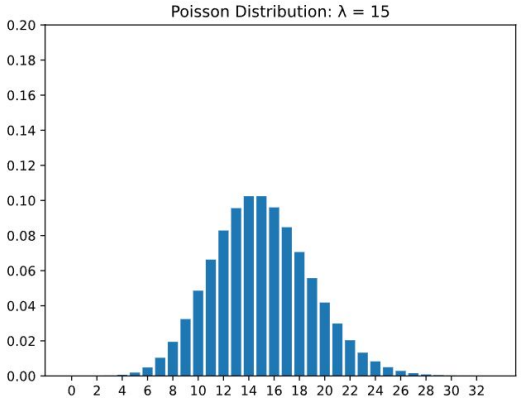


Spread of the Poisson Distribution

- For the Poisson distribution, the variance is simply the value of lambda (λ), meaning that the expected value and variance are equivalent in Poisson distributions

- We know that the Poisson distribution has a discrete random variable and must be greater than 0, so as the expected value increases, the number of possible values the distribution can take on would also increase





Expected Value and Variance of the Binomial Distribution

- Expected (X) = n X p

- Variance (X) = n X p X (1-p)

- Let's say we forgot to study, and we are going to guess B on all 20 questions of a multiple-choice quiz. If we assume that every letter option (A, B, C, and D) has the same probability of being the right answer for each question, how many questions would we expect to get correct?

  Expected(correct answers) = 20*1/4 = 5

- The variance around getting an individual question correct would be p*(1-p), or 0.25*0.75. We then multiply this variance for all 20 questions in the quiz and get:

  Variance (correct answers) = 20 X 0.25 X (1-0.25) = 3.75

Properties of Expectation

- The expected value of two independent random variables is the sum of each expected value separately:
  - $E(X+Y) = E(X) + E(Y)$
  - For example, if we wanted to count the total number of heads between 10 fair gold coin flips and 6 fair silver coin flips, the expected value combined would be 5 heads (from the gold) and 3 heads (from the silver) so 8 heads overall

- Adding a constant a to the distribution changes the expected value by the value a:
  - $E(X+a) = E(X)+a$
  - Let's say that a test was given and graded, and the average grade was 78 out of 100 points. If the teacher decided to curve the grade by adding 2 points to everyone's grade, the average would now be 80 points

Properties of Variance

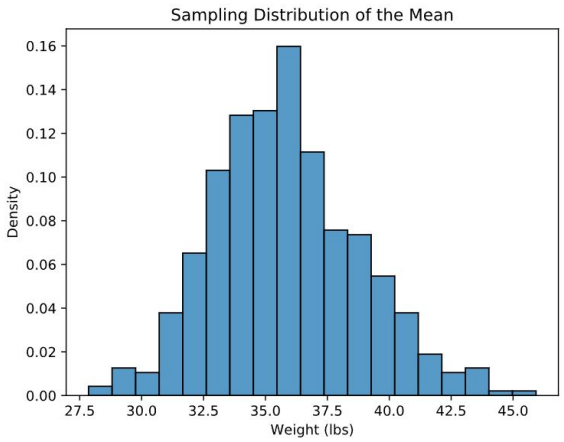
- Increasing the values in a distribution by a constant a does not change the variance:
  - Var(X+a) = Var(X)
  - Let's take the previous example with the teacher curving grades: though the expected value (or average) of the test changes from 78 to 80, the spread and dispersion (or variance) of the test scores stays the same.

- Scaling the values of a random variable by a constant a scales the variance by the constant squared:
  - Var(aX) = a^2 Var(X)
  - Var(X+Y) = Var(X) + Var(Y) if X and Y are independent

Sampling from a Population

- In statistics, we often want to learn about a large population. Since collecting data for an entire population is often impossible, researchers may use a smaller sample of data to try to answer their questions

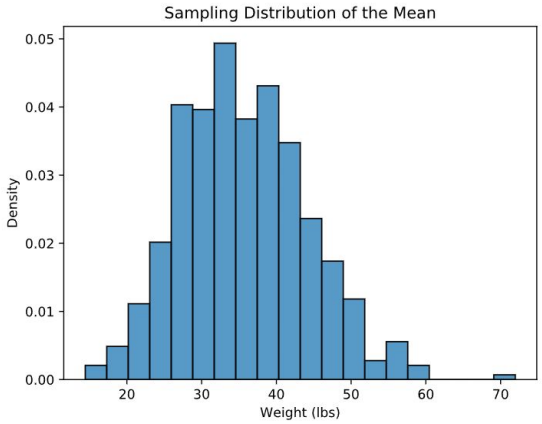
- Each time we sample from a population, we will get a slightly different sample mean. In order to understand how much variation we can expect in those sample means, we can do the following:
  - Take a bunch of random samples with same size
  - Calculate the sample mean for each one
  - Plot a histogram of all the sample means



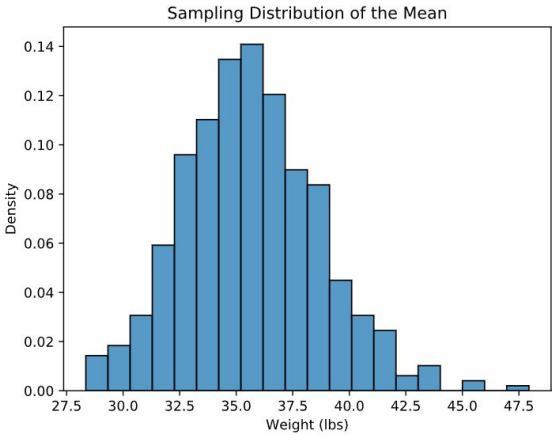
Central Limit Theorem

- We can simulate an approximated sampling distribution for a few different statistics (mean, maximum, variance, etc.). The Central Limit Theorem (CLT) allows us to specifically describe the sampling distribution of the mean

- The CLT states that the sampling distribution of the mean is normally distributed as long as the population is not too skewed or the sample size is large enough



Sample size = 6

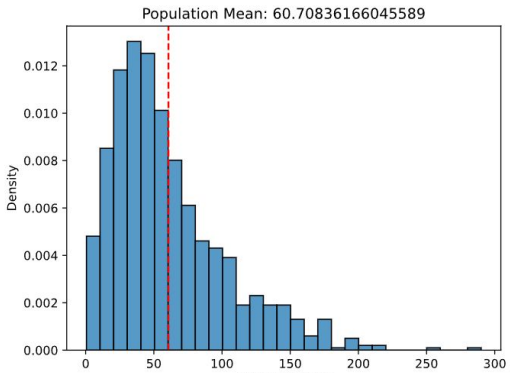


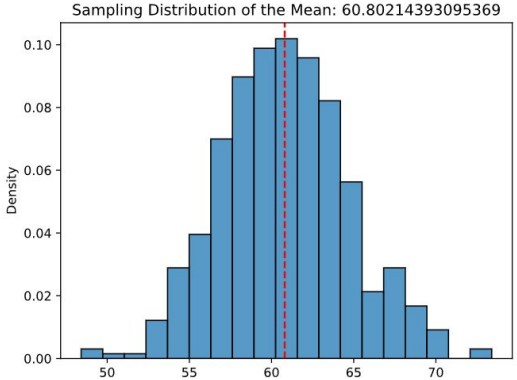
Sample size = 50

Central Limit Theorem

- We take samples of size n from a population (that has a true population mean μ and standard deviation of σ) and calculate the sample mean x

- Given that n is sufficiently large (n > 30), the sampling distribution of the means will be normally distributed with:
  - mean x approximately equal to the population mean μ
  - standard deviation equal to the population standard deviation divided by the square root of the sample size. We can write this out as:
    - Sampling Distribution Standard deviation = $\sigma / \sqrt{n}$

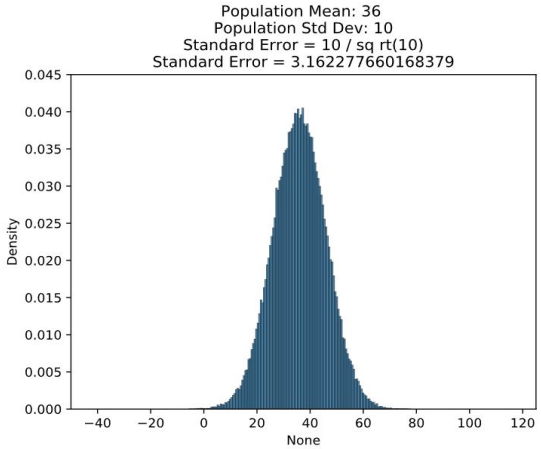


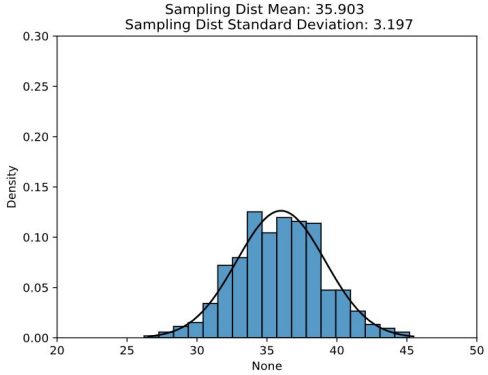


Central Limit Theorem

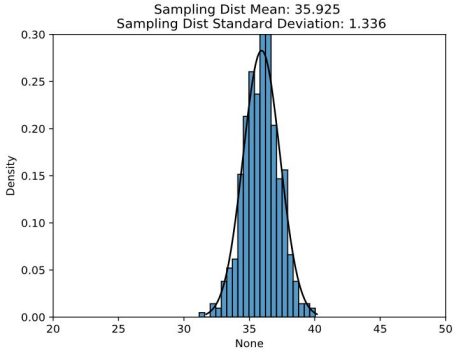
- The standard deviation of a sampling distribution is also known as the standard error of the estimate of the mean

- In many instances, we cannot know the population standard deviation, so we estimate the standard error using the sample standard deviation:
  - Sampling Distribution Standard deviation = standard deviation of sample / √n
  - As sample size increases, the standard error will decrease
  - As the population standard deviation increases, so will the standard error





Sample size = 10



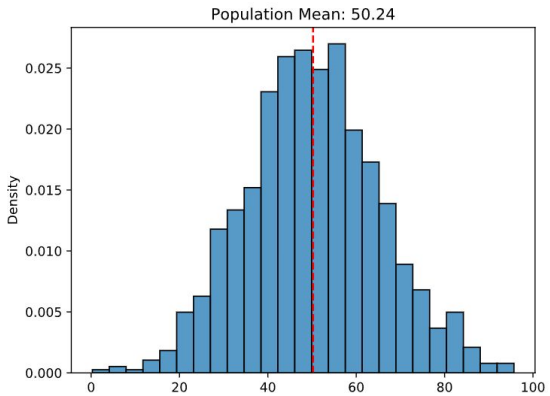
Sample size = 50

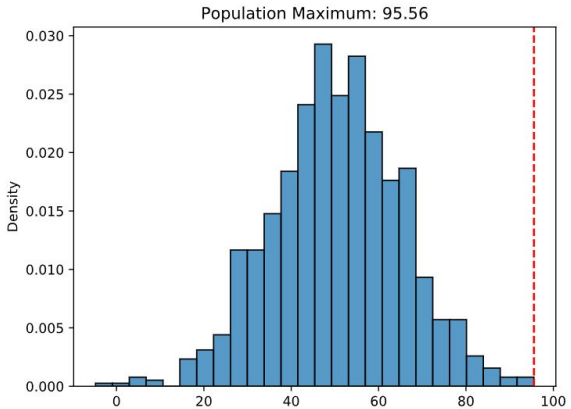
Biased Estimators

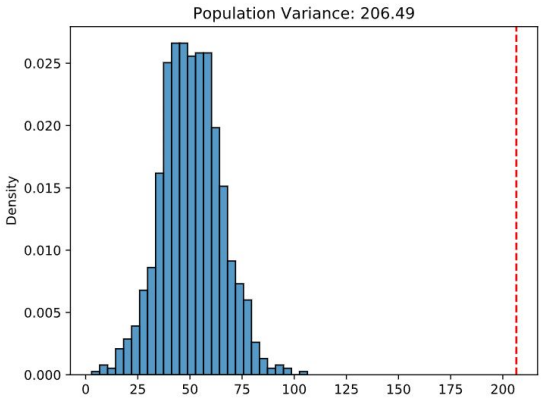
- Because the mean of the sampling distribution of the mean is equal to the mean of the population, we call it an unbiased estimator

- A statistic is called an unbiased estimator of a population parameter if the mean of the sampling distribution of the statistic is equal to the value of the statistic for the population

- The maximum is one example of a biased estimator, meaning that the mean of the sampling distribution of the maximum is not centered at the population maximum







Calculating Probabilities

- Let’s say we are transporting the salmon and want to make sure the crate we carry the fish in will be strong enough to hold the weight:
    - Suppose we estimate that the salmon population has an average weight of 60 lbs with a standard deviation of 40 lbs.
    - We have a box that supports 750 lbs, and we want to be able to transport 10 fish at a time
    - We want to calculate the probability that the average weight of those 10 fish is less than or equal to 75 (750/10)

Calculating Probabilities

- Using the CLT, we first estimate that the mean weight of 10 randomly sampled salmon from this population is normally distributed with mean = 60 and standard error = 40/√10

- Then, we can use this probability distribution to calculate the probability that 10 randomly sampled fish will have a mean weight less than or equal to 75

- Given the probability calculated, would you recommend trying to carry 75 salmon in the box?

Calculating Probabilities with CDF

```
import scipy.stats as stats
x = 75
mean = 60
std_dev = 40
samp_size = 10
standard_error = std_dev / (samp_size**.5)
stats.norm.cdf(x,mean,standard_error)
```

```
0.8821600432854813
```

Calculating Probabilities

- In real life, the data scientist is often stuck with their one sample mean as a best guess for the population mean. However, they can leverage the CLT to estimate the standard error

- While a researcher or data scientist probably does not know the population standard deviation, they can use the standard deviation of their sample to estimate it

- Leveraging the part of the CLT that says the sampling distribution is normally distributed, our data scientist can use a nifty property of normal distributions: 95% of normally distributed values are within about 1.96 standard deviations of the mean. This allows the data scientist to estimate the width of the sampling distribution

Building intuition for the CLT

- Assume a data scientist collected a single sample of 150 wages and calculated an average wage of 17.74 dollars
- The data scientist needs to multiply 1.96 by the estimated standard error: 1.96 * 1.275 = 2.50
- 95% of those samples would be within about 2.50 dollars from the true population mean
- Therefore, there is about a 95% probability that the observed sample mean of 17.74 is no more than 2.50 dollars away from the population mean. In other words, there is about a 95% probability that the population mean is between 15.24 and 20.24. This is referred to as a 95% confidence interval.

```
std_error = np.std(my_sample)/(150 ** 0.5)
print(std_error)
# output: 1.275
```

Descriptive statistics vs Inferential Statistics

- Descriptive statistics **can tell us about the data that we have**, but sometimes we can't collect all of the data that we need to answer our questions.

- For example, maybe we want to know whether people who get a vaccine are less likely to get a disease. We can't vaccinate every single person in the world to test this, so we'll have to vaccinate a smaller sample of people

- If the vaccine seems to work in our sample, we need to know whether that could have been a random result - This is where hypothesis testing can help

Descriptive statistics vs Inferential Statistics

- Inferential statistics is all about **using a sample** (a subset of a population) **to make inferences about a larger population of interest**.

- Suppose you are a researcher studying the relationship between high school students' homework grades and standardized test scores

- It would be very difficult and expensive to collect information on homework grades and standardized test scores for every single high school student in the world.

- Instead, you could find a random sample of students and inspect the relationship between homework grades and standardized test scores among that sample

- Finally, you could use a regression analysis to understand whether a similar relationship is likely to exist in the larger population of all students

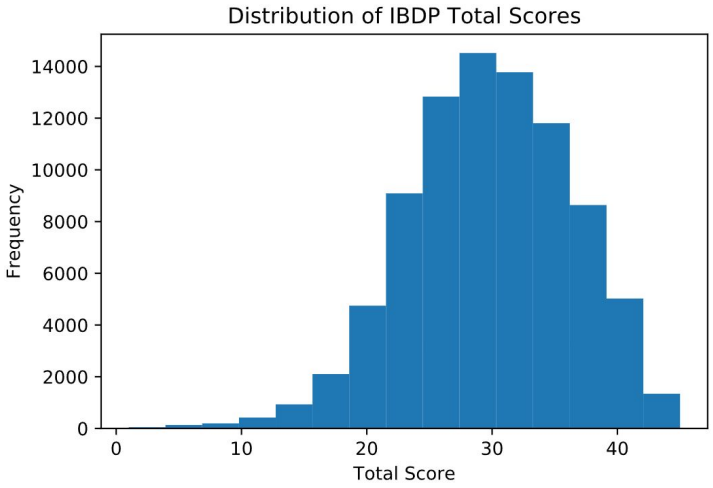
What is hypothesis testing?

- Hypothesis Testing is a framework for asking questions about a dataset and answering them with probabilistic statements. There are many different kinds of hypothesis tests that can be used to address different kinds of questions and data:
    - Step 1: Ask a Question
    - Step 2: Define the Null and Alternative Hypotheses
    - Step 3: Determine the Null Distribution
    - Step 4: Calculate a P-Value (or Confidence Interval)
    - Step 5: Interpret the Results

Hypothesis testing - Case Study

- The International Baccalaureate Degree Programme (IBDP) is a world-wide educational program. Each year, students in participating schools can take a standardized assessment to earn their degree

- Total scores on this assessment range from 1-45. Based on data provided here, the average total score for all students who took this exam in May 2020 was around 29.92



Hypothesis testing - Ask a Question

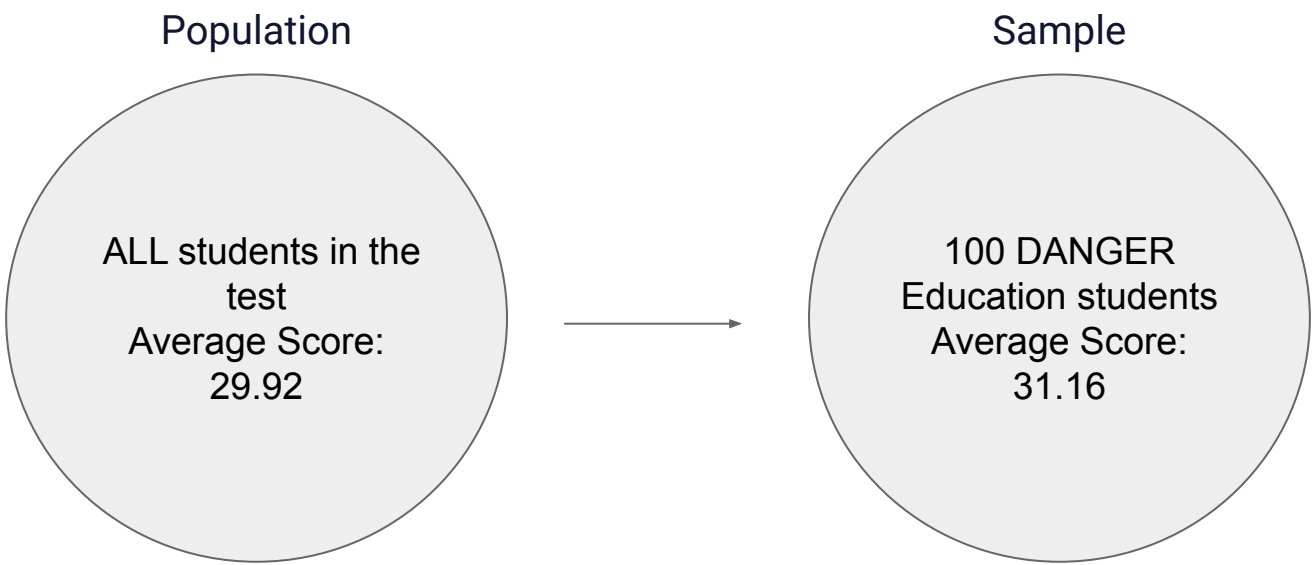
- Imagine DANGER Education offers test-preparation program. Suppose that 100 students who took the IBDP assessment in May 2020 were randomly chosen to participate in the first group of this program and that these 100 students earned an average score of 31.16 points on the exam — about 1.24 points higher than the international average

- Are these students really outperforming their peers due to DANGER Education program?

- The trick is to quantify when differences are “large enough” to convince us that these students are systematically different from the general population.

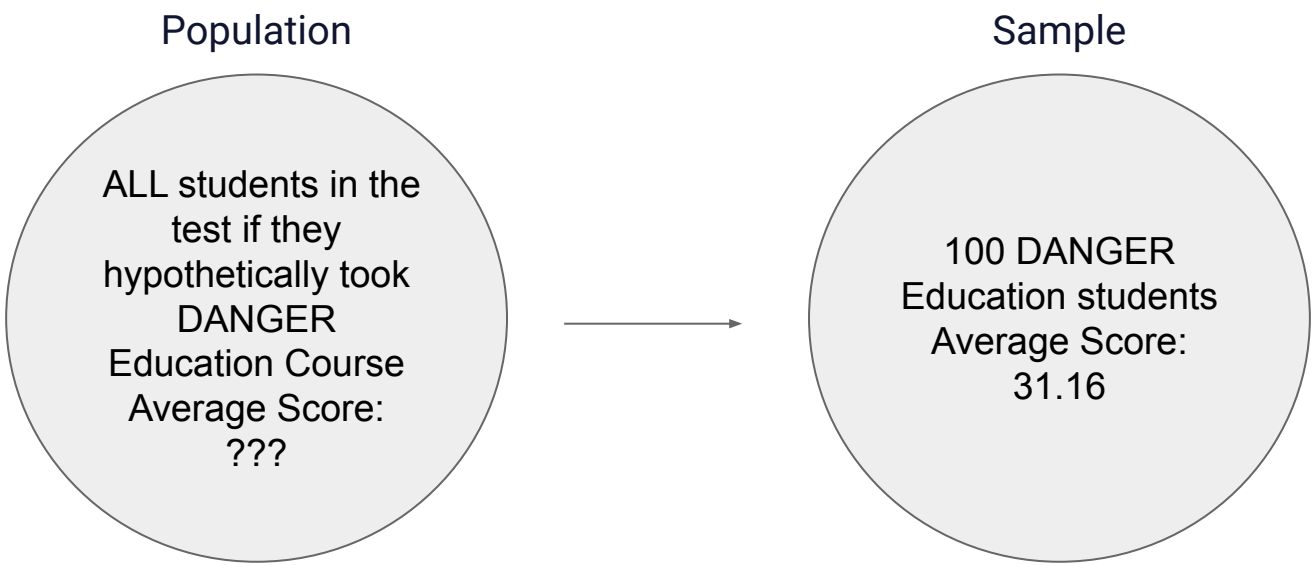
Hypothesis testing - Define the Null and Alternative Hypotheses

- Hypothesis 1 (the Null Hypothesis): The 100 DANGER Education students are a random sample from the general population of test takers, who had an average score of 29.92. If this hypothesis is true, DANGER Education students earned a slightly higher average score by **random chance**



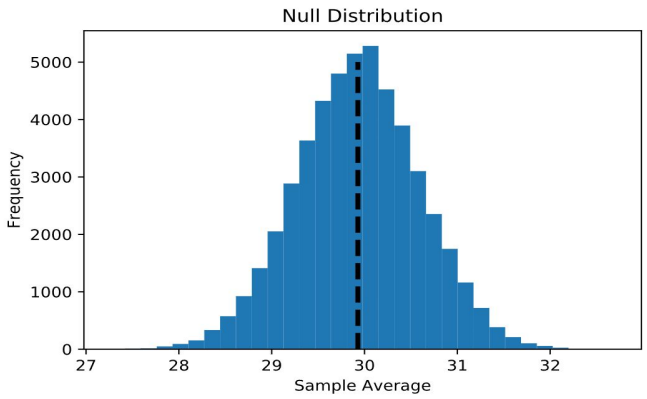
Hypothesis testing - Define the Null and Alternative Hypotheses

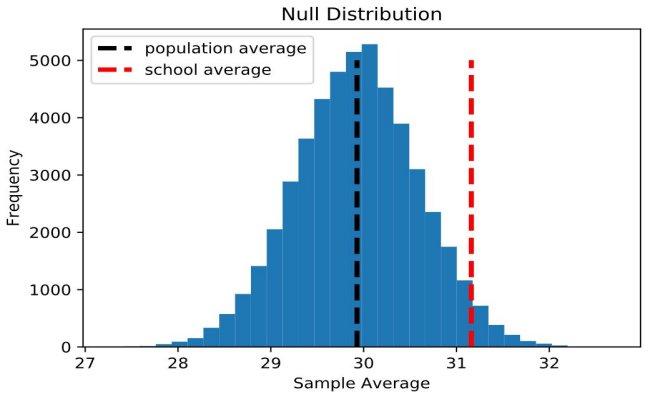
- Hypothesis 2 (the Alternative Hypothesis): The 100 DANGER Education students came from a population with an average score that is different from 29.92. If the alternative hypothesis is true, our sample of 100 DANGER Education came from a **different population than the other test-takers.**



Hypothesis testing - Determine the Null Distribution

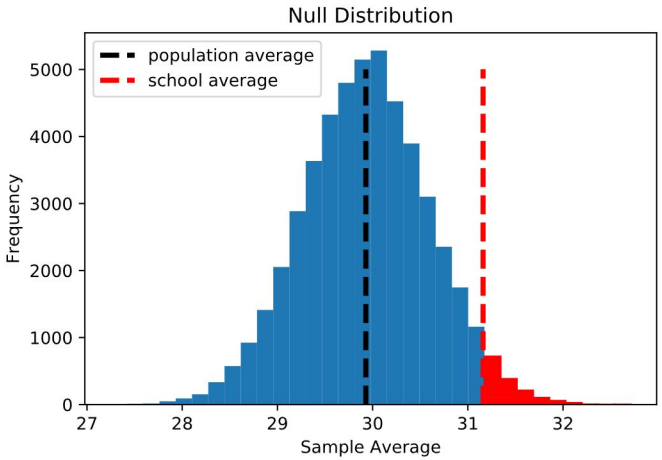
- Now that we have our null hypothesis, we can generate a null distribution: the distribution (across repeated samples) of the statistic we are interested in if the null hypothesis is true. In the IBDP example described above, this is the distribution of average scores for repeated samples of size 100 drawn from a population with an average score of 29.92:
    - Taking many random samples of 100 scores, each, from the population
    - Calculating and storing the average score for each of those samples
    - Plotting a histogram of the average scores
- If the null hypothesis is true, then the average score of 31.16 observed among DANGER Education students is simply one of the values from this distribution.





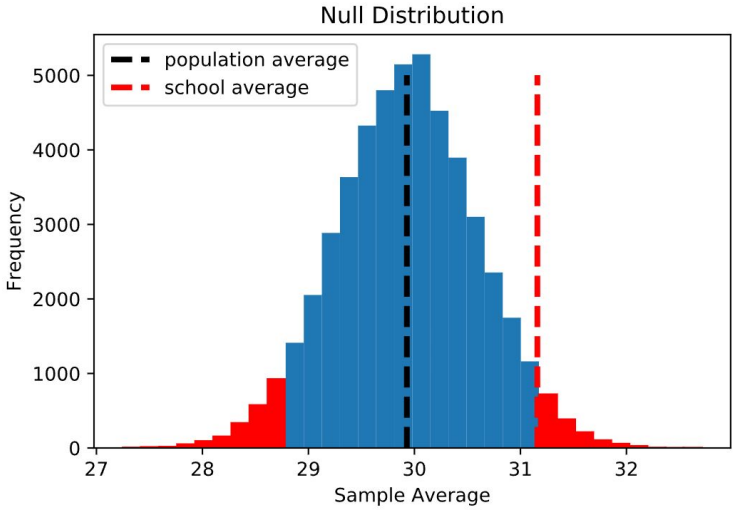
Hypothesis testing - Calculate a P-Value (or Confidence Interval)

- Given that the null hypothesis is true (that the 100 students from DANGER Education were sampled from a population with an average IBDP score of 29.92), how likely is it that their average score is 31.16?
- Option 1 - we want to know the probability of observing a sample average greater than or equal to 31.16 given that the null hypothesis is true.
- The red region makes up about 3.1% of the total distribution. This proportion, which is usually written in decimal form (i.e., 0.031), is called a p-value.



Hypothesis testing - Calculate a P-Value (or Confidence Interval)

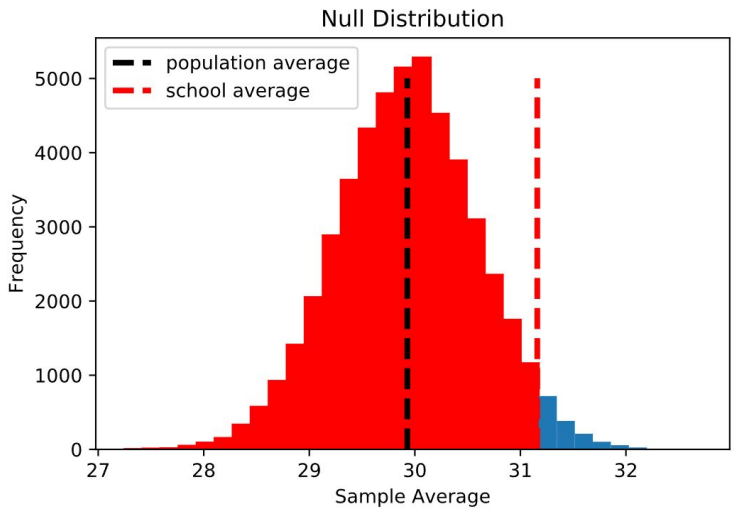
- Option 2 - we're interested in the probability of observing a sample average that is at least 1.24 points DIFFERENT (higher OR lower) than the population average.
- This area is twice as large as in the previous example, leading to a p-value that is also twice as large: 0.062.
- This version is referred to as a Two Sided or Two-Tailed test



Hypothesis testing - Calculate a P-Value (or Confidence Interval)

- Option 3 - We want to know the probability of observing a sample average less than or equal to 31.16, given that the null hypothesis is true

- This is the proportion of the null distribution that is less than or equal to 31.16. This is a very large area, leading to a p-value of 0.969.



Hypothesis testing - Interpret the Results

- Consider the first p-value of 0.031. The interpretation of this number is as follows:
    - If the 100 students at DANGER Education were randomly selected from the full population (which had an average score of 29.92), there is a 3.1% chance of their average score being 31.16 points or higher

- This means that it is relatively unlikely that the DANGER Education students scored higher (on average) than their peers by random chance, despite no real difference at the population level.

- In other words, the observed data is unlikely if the null hypothesis is true.

- Many data scientists use a predetermined threshold to decide whether a particular p-value is significant or not - A common choice for this threshold, which is also sometimes referred to as Alpha, is 0.05

Assumption of test

- When running any hypothesis test, it is important to know and verify the assumptions of the test:
    - The sample was randomly selected from the population
    - The individual observations were independent
    - The data is normally distributed without outliers
      (e.g. good idea to make sure to plot a histogram of your data)

Introduction to Significance Thresholds

- Sometimes, when we run a hypothesis test, we simply report a p-value or a confidence interval

- In other situations, we want to use our p-value to make a decision or answer a yes/no question

- For example, suppose that we're developing a new quiz question and want learners to have a 70% chance of getting the question right. We show our quiz question to a sample of 100 learners and 60 of them get it right. Is this significantly different from our target of 70%? If so, we might try to rewrite it.

- In order to turn a p-value, which is a probability, into a yes or no answer, data scientists often use a pre-set significance threshold

- The significance threshold can be any number between 0 and 1, but a common choice is 0.05.

- P-values that are less than this threshold are considered "significant", while larger p-values are considered "not significant".

Interpreting a P-Value based on a Significance Threshold

- For example:
  - Null: The probability that a learner gets the question correct is 70%
  - Alternative: The probability that a learner gets the question correct is **not** 70%.

- If the p-value is less than 0.05, the p-value is significant. We will “reject the null hypothesis” and conclude that the probability of a correct answer is significantly different from 70%. This would lead us to re-rewrite the question.

- If the p-value is greater than 0.05, the p-value is not significant. We will not be able to reject the null hypothesis, and will conclude that the probability of a correct answer is not significantly different from 70%. This would lead us to leave the question on the site.

Error Types

- Whenever we run a hypothesis test using a significance threshold, we expose ourselves to making two different kinds of mistakes: type I errors (false positives) and type II errors (false negatives)

Null Hypothesis	Is True	Is False
P-Value Significant	Type 1 Error	Correct
P-Value Not Significant	Correct	Type 2 Error

Error Types

- Suppose that the average score on a standardized test is 50 points. A researcher wants to know whether students who take this test in a larger classroom score significantly differently from the general population of test-takers. The researcher randomly assigns 100 students to take the test in a larger classroom. Then, the researcher runs a hypothesis test with a significance threshold of 0.05
- Null: The mean score for students who take the test in a larger classroom is 50 points.
- Alternative: The mean score for students who take the test in a larger classroom is not 50 points.
- Suppose that the truth (which the researcher doesn't know) is: if every student took the test in a larger classroom, the average score for all test-takers would be 52 points.
- Based on their sample of only 100 students, the researcher calculates a p-value of 0.07
- **Does the researcher make any mistake? If yes what type of error does he makes?**

Setting the Type I Error Rate

- It turns out that, when we run a hypothesis test with a significance threshold, the **significance threshold** is equal to the **type I error** (false positive) rate for the test. To see this, we can use a simulation:
  - Simulate 100 learners, where each learner has a 70% chance of answering a quiz question correctly.
  - Calculate the number of simulated learners who answered the question correctly. Note that, because each learner has a 70% chance of answering correctly, this number will likely be around 70, but will vary every time by random chance.
  - Run a binomial test for the simulated sample where the null hypothesis is that the probability of a correct answer is 70% (0.7). Note that, every time we run this test, the null hypothesis is true because we simulated our data so that the probability of a correct answer is 70%.
  - Add 1 to our false positives counter every time we make a type I error (the p-value is significant)

- Note that the proportion of false positive tests is very similar to the value of the significance threshold (0.05).

Problems with Multiple Hypothesis Tests

- While significance thresholds allow a data scientist to control the false positive rate for a single hypothesis test, this starts to break when performing multiple tests as part of a single study

- If the null hypothesis is true for every hypothesis test (the probability of a correct answer is 70% for every question) and we use a .05 significance level for each test

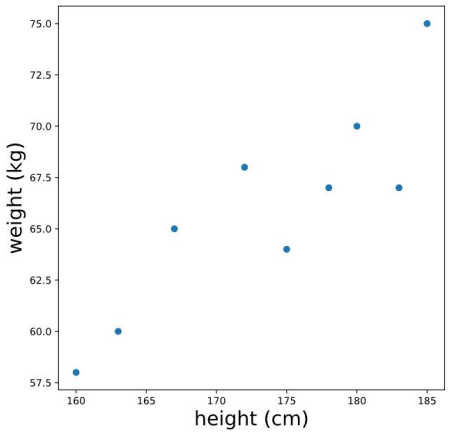
- When we run a hypothesis test for a single question, we have a 95% chance of getting the right answer (a p-value > 0.05) — and a 5% chance of making a type I error

- When we run hypothesis tests for two questions, we have only a 90% chance of getting the right answer for both hypothesis tests (.95*.95 = 0.90) — and a 10% chance of making at least one type I error.

Introduction to Linear Regression

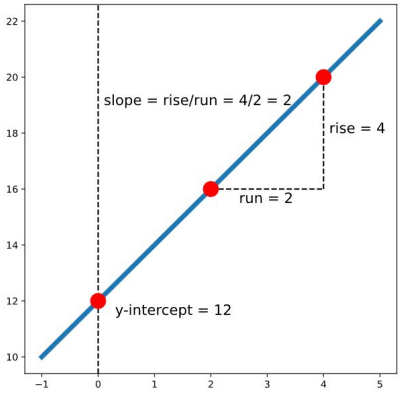
- Linear regression is a powerful modeling technique that can be used to understand the relationship between a quantitative variable and one or more other variables, sometimes with the goal of making predictions. For example, linear regression can help us answer questions like:
  - What is the relationship between apartment size and rental price for HK apartments?
  - Is a father's height a good predictor of their child's adult height?

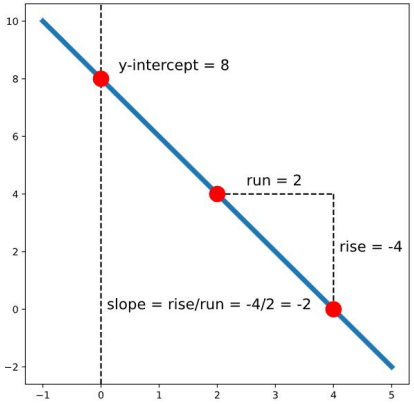
- The first step before fitting a linear regression model is exploratory data analysis and data visualization: **is there a relationship that we can model?**



Equation of a Line

- Like the name implies, Linear regression involves fitting a line to a set of data points. In order to fit a line, it's helpful to understand the equation for a line, which is often written as y=mx+b. In this equation:
  - x and y represent variables, such as height and weight or hours of studying and quiz scores
  - b represents the y-intercept of the line. This is where the line intersects with the y-axis (a vertical line located at x = 0)
  - m represents the slope. This controls how steep the line is



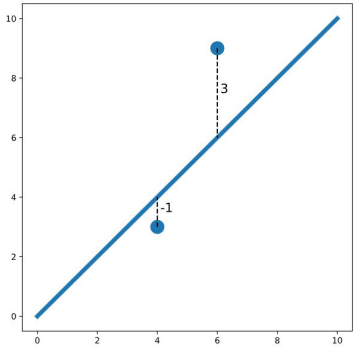


Finding the "Best" Line

- A common choice for linear regression is ordinary least squares (OLS). In simple OLS regression, we assume that the relationship between two variables x and y can be modeled as:
  - y = mx + b + error

- We define “best” as the line that minimizes the total squared error for all data points. This total squared error is called the loss function in machine learning.

- Notice that we square each individual distance so that points below and above the line contribute equally to loss (when we square a negative number, the result is positive). To find the best-fit line, we need to find the slope and intercept of the line that minimizes loss.



Fitting a Linear Regression Model in Python

- There are a number of Python libraries that can be used to fit a linear regression. One of them is the OLS.from_formula() function from statsmodels.api, which uses simple syntax and provides comprehensive model summaries (test_data.csv)

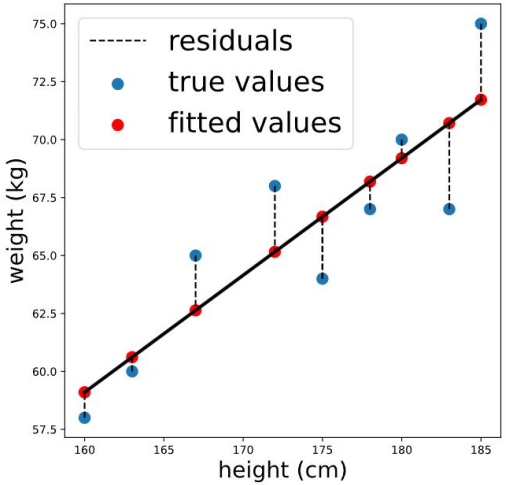
- The regression model allows us to predict the dependent variable (y) with the independent variable (x)

- Note that the slope gives us two pieces of information: the magnitude AND the direction of the relationship between the x and y variables.

Assumptions of Linear Regression (1)

- The relationship between the outcome variable and predictor is **linear** (can be described by a line). We can check this before fitting the regression by simply looking at a plot of the two variables

- The fitted values are the predicted value for the dataset which was used to fit the model, while the residuals are the differences between the predicted value and the true value

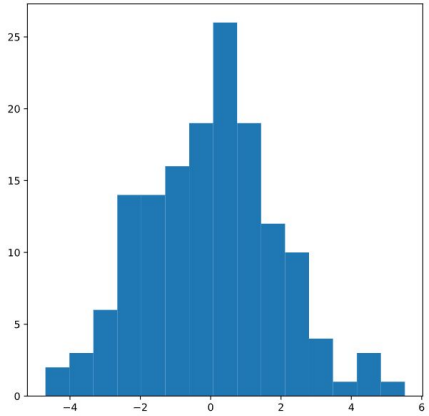


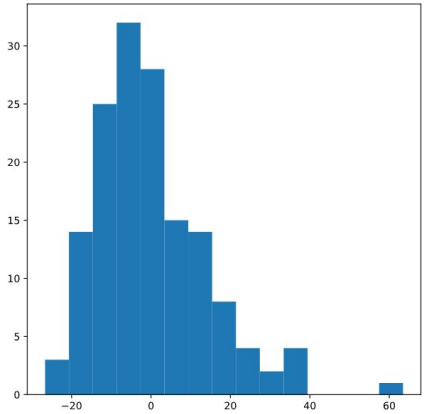
Assumptions of Linear Regression (2)

- The **normality assumption** states that the residuals should be normally distributed. To check this assumption, we can inspect a histogram of the residuals and make sure that the distribution looks approximately normal

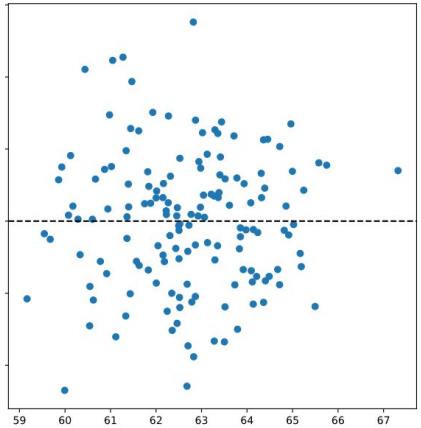
- **Homoscedasticity** is saying that the residuals have equal variation across all values of the predictor variable. A common way to check this is by plotting the residuals against the fitted values.

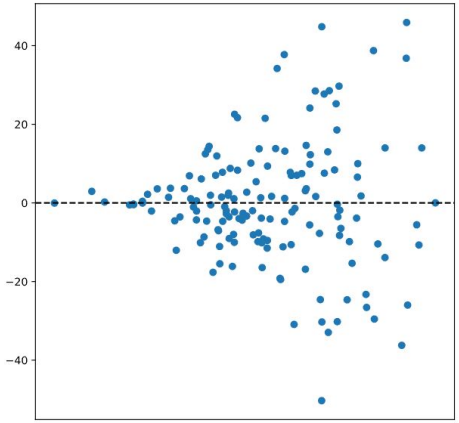
Normal vs Skewed





Homoscedasticity vs Heteroskedasticity

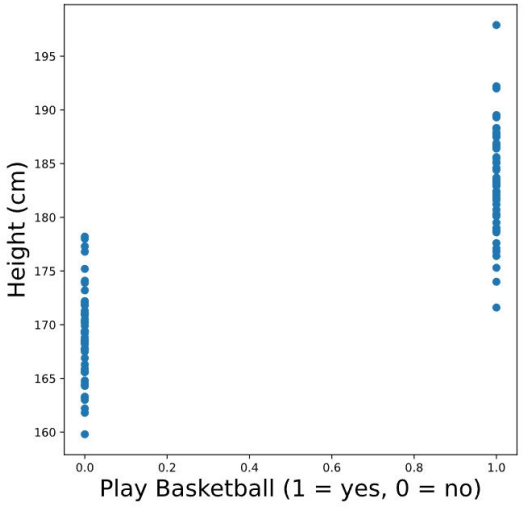


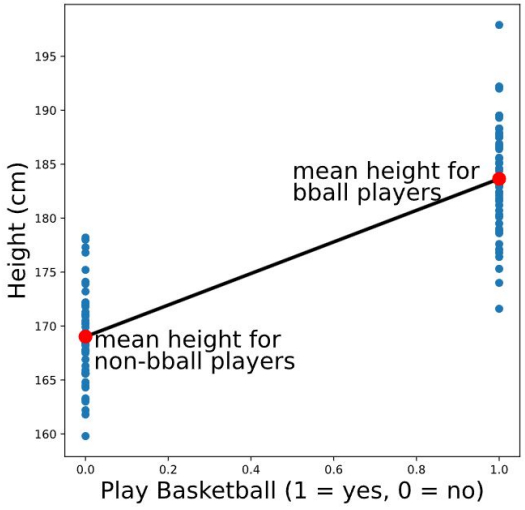


Categorical Predictors

- We can also use categorical predictors. The simplest case of a categorical predictor is a binary variable (only two categories)

- For example, suppose we surveyed 100 adults and asked them to report their height in cm and whether or not they play basketball. We've coded the variable bball_player so that it is equal to 1 if the person plays basketball and 0 if they do not





Unit 3
Exploratory Data Analysis

What will you learn

- Basic summary statistics and visualizations
- Exploring associations between Quantitative variables and Categorical variables
- Exploring associations between two Quantitative variables
- Exploring associations between two Categorical variables

What is EDA?

- Exploratory Data Analysis (EDA for short) is all about getting curious about your data – finding out what is there, what patterns you can find, and what relationships exist

- EDA is the important first step towards analysis and model building. When done well, it can help you formulate further questions and areas for investigation

- The main goals of EDA are generally
  - Uncover the data structure and determine how it is coded
  - Inspect and “get to know” the data by summarizing and visualizing it
  - Detect outliers, missing data, and other anomalies and decide how/whether to address these issues
  - Find new avenues for analysis and further research
  - Prepare for model building or analysis, including the following:
    - Check assumptions
    - Select features
    - Choose an appropriate method

EDA techniques - Numerical Summaries

- Numerical summaries are a great way to condense the information we have into a more reasonable amount of space. For numerical data, this allows us to get a sense of scale, spread, and central tendency

- In Pandas, we can use dataframe.describe() to summarize the central tendency, dispersion and shape of a dataset's distribution

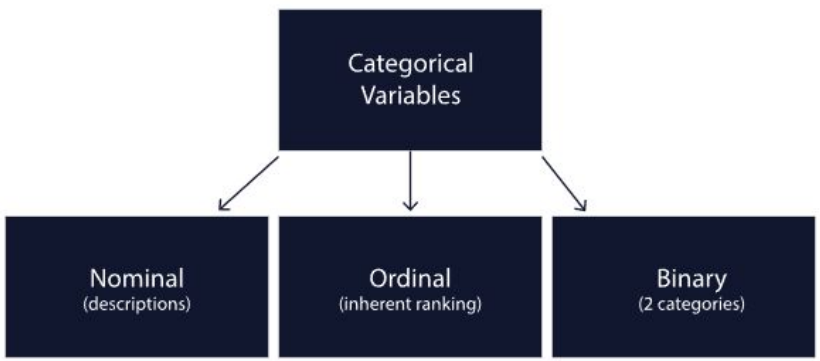
```
>>> df.describe(exclude=[object])
       categorical  numeric
count            3      3.0
unique           3      NaN
top              f      NaN
freq             1      NaN
mean           NaN      2.0
std            NaN      1.0
min            NaN      1.0
25%            NaN      1.5
50%            NaN      2.0
75%            NaN      2.5
max            NaN      3.0
```

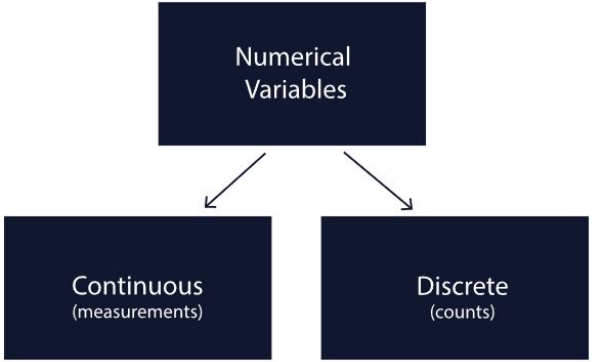
EDA techniques - Visual Summaries

- Visual summaries can provide even more context and detail in a small amount of space.
- There are many different types of visualizations that we might want to create as part of EDA. For example, histograms allow us to inspect the distribution of a quantitative feature, providing information about central tendency, spread, and shape (eg., skew or multimodality)
- The histogram below shows the distribution of the number of characters in each student's name

Assessing Variable Types - Numerical

- Best practices for data storage say that we should match the data type of the column with its real-world variable type. For example,
  - Continuous (numerical) variables should usually be stored as the **float data type** because they allow us to store decimal values.
  - Discrete (numerical) variables should be stored as the **int datatype** to represent mathematically that they are discrete.





Assessing Variable Types - Categorical

- Just like with numerical variables, best practices for categorical data storage say that we should match the data type of the column with its real-world variable type:
    - Nominal variables are often represented by the **object data type**. Columns in the object data type can contain any combination of values, including strings, integers, booleans, etc
    - Nominal variables are also represented by the **string data type**. However, Pandas usually guesses object rather than string, so if you want a column to be a string, you will likely have to explicitly tell pandas to make it a string. This is most important if you want to do string manipulations on a column like .lower()
    - Ordinal variables should be represented as **objects**, but pandas often guesses int since they are often encoded as whole numbers.
    - Binary variables can be represented as **bool**, but pandas often guesses int or object data types.

Assessing Variable Types - Categorical Order

- For ordinal categorical variables, we often want to store two different pieces of information: category labels and their order

- There is a specific data type for categorical variables in pandas called category. The pandas .Categorical() method can be used to store data as type category and indicate the order of the categories.

Assessing Variable Types - OHE

- Another way of encoding categorical variables is called One-Hot Encoding (OHE). With OHE, we essentially create a new binary variable for each of the categories within our original variable. This technique is useful when managing nominal variables because it encodes the variable without creating an order among the categories.

- It is important to note that OHE works best when we do not create too many additional variables, as increasing the dimensionality of our dataframe can create problems when working with certain machine learning models.

Summarizing Quantitative Data

- Basic summary statistics and visualizations are important components of EDA as they allow us to condense a large amount of information into a small set of numbers or graphics that can be easily interpreted.

- The pandas library offers a handy method .describe() for displaying some of the most common summary statistics for the columns in a DataFrame.

- For quantitative variables, we often want to describe the central tendency, or the “typical” value of a variable - mean, median, mode and trimmed mean

The Spread for Quantitative Data

- The spread of a quantitative variable describes the amount of variability. This is important because it provides context for measures of central tendency.

- There are several common measures of spread:
  - Range: The difference between the maximum and minimum values of a variable.
  - Interquartile range (IQR): The difference between the 75th and 25th percentile values.
  - Variance: The average of the squared distance from each data point to the mean.
  - Standard deviation (SD): The square root of the variance.

- For quantitative variables, boxplots and histograms are two common visualizations. These plots are useful because they simultaneously communicate information about minimum and maximum values, central location, and spread. Histograms can additionally illuminate patterns that can impact an analysis (e.g., skew or multimodality)

- Python's seaborn library, built on top of matplotlib, offers the boxplot() and histplot() functions to easily plot data from a pandas DataFrame

Summarizing Categorical Variables

- A good way to summarize categorical variables is to generate a frequency table containing the count of each distinct value
- Sometimes it is useful to also look at the proportion of values in each category
- For categorical variables, bar charts and pie charts are common options for visualizing the count (or proportion) of values in each category. They can also convey the relative frequencies of each category.
- Python's seaborn library offers several functions that can create bar charts. The simplest for plotting the counts is countplot()
- The pandas library provides a convenient wrapper function around matplotlib's pie() function that can generate a pie chart from any column in a DataFrame

Explore the relationship between variables

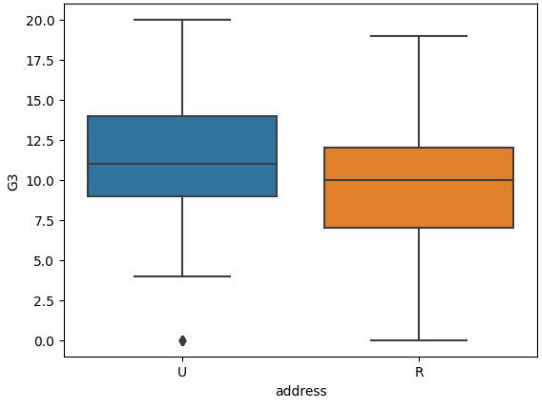
- Examining the relationship between variables can give us key insight into our data

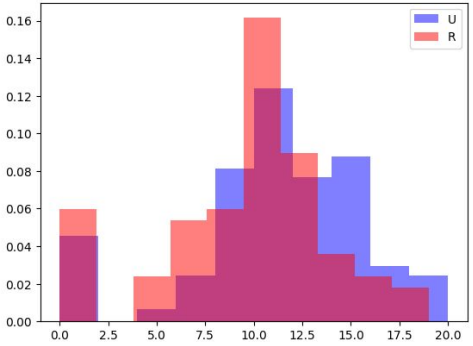
- Suppose that we want to know whether there is an association between student math scores (G3) and the student's address (urban or rural), We can separate out G3 scores into two separate lists: one for students who live in an urban location ('U') and one for students who live in a rural location ('R')

- We can begin quantifying this association by using two common summary statistics, mean and median differences. Highly associated variables tend to have a large mean or median difference

- One way to get a better sense of spread is by looking at a visual representation of the data. Side-by-side box plots are useful in visualizing mean and median differences because they allow us to visually estimate the variation in the data

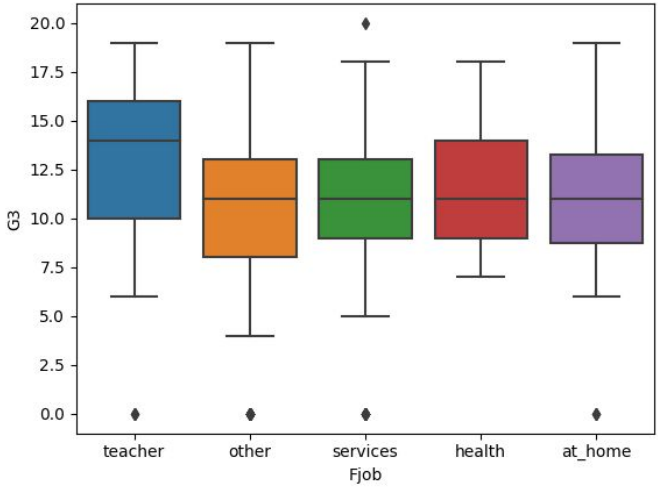
- Another way to explore the relationship between a quantitative and categorical variable in more detail is by inspecting overlapping histograms. IF there is also still a lot of overlap, it is suggesting that the association is relatively weak





Exploring Non-Binary Categorical Variables

- When looking at an association between a quantitative variable and a non-binary categorical variable, we must examine all pair-wise differences
- The easiest way to quickly visualize these comparisons is with side-by-side box plots
- Visually, we need to compare each box to every other box. While most of these boxes overlap with each other, there are some pairs for which there are some apparent differences.



Explore relationship between two quantitative variables

- One of the best ways to quickly visualize the relationship between quantitative variables is to plot them against each other in a scatter plot
- It's important to note that different kinds of associations can lead to different patterns
- Covariance is a summary statistic that describes the strength of a linear relationship.
- A positive covariance indicates that a larger value of one variable is associated with a larger value of the other. A negative covariance indicates a larger value of one variable is associated with a smaller value of the other. A covariance of 0 indicates no linear relationship.

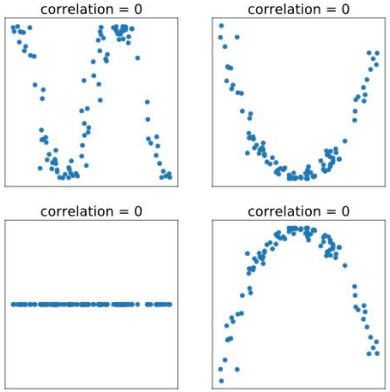
	variable 1	variable 2
variable 1	variance(variable 1)	covariance
variable 2	covariance	variance(variable 2)

Explore relationship between two quantitative variables

- Correlation is a scaled form of covariance. It also measures the strength of a linear relationship, but ranges from -1 to +1, making it more interpretable.

- Highly associated variables with a positive linear relationship will have a correlation close to 1. Highly associated variables with a negative linear relationship will be close to -1

- Because correlation and covariance both measure the strength of linear relationships with non-zero slopes, but not other kinds of relationships, correlation can be misleading



Explore relationship between two categorical variables

- Contingency tables, also known as two-way tables or cross-tabulations, are useful for summarizing two variables at the same time

- To assess whether there is an association between these two variables, we need to ask whether information about one variable gives us information about the other.

- Sometimes it's helpful to convert those frequencies to proportions. We can accomplish this simply by dividing the all the frequencies in a contingency table by the total number of observations

- The proportion of respondents in each category of a single question is called a marginal proportion

```
leader              influence
no      0.484       no      0.388
yes     0.516       yes     0.612
```

Explore relationship between two categorical variables

- In order to understand whether these questions are associated, we can use the marginal proportions to create a contingency table of expected proportions if there were no association between these variables

- To calculate these expected proportions, we need to multiply the marginal proportions for each combination of categories. The more that the expected and observed tables differ, the more sure we can be that the variables are associated.

	leader = no	leader = yes
influence = no	0.484*0.388 = 0.188	0.516*0.388 = .200
influence = yes	0.484*0.612 = 0.296	0.516*0.612 = 0.315

	leader = no	leader = yes
influence = no	0.188*11097 = 2087	0.200*11097 = 2221
influence = yes	0.296*11097 = 3288	0.315*11097 = 3501

leader	no	yes
influence		
no	3015	1293
yes	2360	4429

The Chi-Square Statistic

- Many data scientists use the Chi-Square statistic to summarize how different these two tables are

- To calculate the Chi Square statistic, we simply find the squared difference between each value in the observed table and its corresponding value in the expected table, and then divide that number by the value from the expected table; finally add up those numbers:

$$ChiSquare = \sum \frac{(observed - expected)^2}{expected}$$

- The interpretation of the Chi-Square statistic is dependent on the size of the contingency table. For a 2x2 table (like the one we've been investigating), a Chi-Square statistic larger than around 4 would strongly suggest an association between the variables